ETH zürich

Integrated Systems Laboratory



IHP 130 nm SG13G2 EZ Standard-Cell Library 8-Track v1.1

DATABOOK

TT | 1.20 V | 25 °C

Authors

The EZ130 8T library was created as part of the "VLSI 5: Design of a Standard-Cell Library" course from ETH Zurich during its first, spring 2025 offering. Its contributors are:

Oscar Castañeda, Lavinia Recchioni, Tobias Senti, Flurin Cahenzli, Marco Ferroni, Thorben Heekenjann, Robert Kenter, Stefan Odermatt, Daniel Richner, Gabriel Altendorfer, Davide Cannone, Miguel Correa, Ivan Herger, Lars Kröger, Nicolas Nanzer, Darja Nonaca, Christopher Reinwardt, Lukas Winklhofer, Enrico Zelioli, Yiheng Zhang, Yingxue Zhang, Zerun Jiang, Beat Muheim, Arianna Rubino, Jérémy Guichemerre, Frank K. Gürkaynak, and Christoph Studer.

If you use the EZ130 8T library (or parts of it), then you must cite this reference:

```
@article{castaneda2026ez130t8,
  author = {Casta\~neda, Oscar and Recchioni, Lavinia and Senti, Tobias and
      Cahenzli, Flurin and Ferroni, Marco and Heekenjann, Thorben and Kenter,
      Robert and Odermatt, Stefan and Richner, Daniel and Altendorfer, Gabriel and
       Cannone, Davide and Correa, Miguel and Herger, Ivan and Kr\"oger, Lars and
      Nanzer, Nicolas and Nonaca, Darja and Reinwardt, Christopher and Winklhofer,
       Lukas and Zelioli, Enrico and Zhang, Yiheng and Zhang, Yingxue and Keller,
      Domenic and Mirfarshbafan, Seyed Hadi and Jiang, Zerun and Muheim, Beat and
      Rubino, Arianna and Guichemerre, J\'er\'emy and G\"urkaynak, Frank K. and
      Studer, Christoph},
  title = {An Open-Source Standard-Cell Library for {IHP} 130\,nm Developed by
      Students},
  journal = {arXiv preprint arXiv:XXXX.XXXXX},
  month = {August},
  year = {2026},
  url = {https://arxiv.org/abs/????}
}
```

License

The EZ130 8T library is provided as is under the terms of the Apache 2.0 license.

Resources

The most recent version of the EZ130 8T library is available at `https://ez.ethz.ch`.

Acknowledgments

The EZ130 8T library authors thank Cadence Designs Systems for their generous support by providing licenses to design and characterize the standard-cell library.

Contents

Physical Specifications 4
Characterization Parameters 4
AND2 5
AND3 7
AND3B 9
AND4 11
ANTENNA 13
AO21 14
AO211 15
AO22 17
AO31 19
AOI21 21
AOI2B1 23
AOI21B 25
AOI211 27
AOI22 29
AOI2B2 31
AOI2BB2 33
AOI2B2B 35
AOI221 37
AOI31 39
AOAI211 41
BUF 43
CG 46
CGTS 48
DECAP 50
DFFQ 51
DFFR 54
DFFRQ 57
DFFRQ2 60
DFFRQ4 62
DFFMRQ 65
DLY1 67
FA 68
HA 70
INV 71
LATQ 74
LATR 76
LATNQ 78
LATNR 80

MIN3 82
MUX2 84
MUXI2 86
MUX4 88
NAND2 90
NAND2B 92
NAND3 94
NAND3B 96
NAND4 98
NAND4B 99
NAND4BB 101
NOR2 103
NOR2B 105
NOR3 107
NOR3B 109
NOR4 111
NOR4B 113
NOR4BB 115
OA21 117
OA211 118
OAI21 120
OAI211 122
OAI22 124
OAI2BB2 126
OAI2B2B 127
OAI221 129
OAI31 131
OAOI211 133
OR2 135
OR3 137
OR3B 139
OR4 141
TBUF 143
TIEHI 145
TIELO 146
TINV 147
XNOR2 149
XNOR3 151
XOR2 153
XOR3 155

Physical Specifications

Metal Layers	1
Drawn Gate Length [µm]	0.130
Layout Grid [µm]	0.005
Vertical Pin Grid [µm]	0.42
Horizontal Pin Grid [µm]	0.42
X-Coordinate Pin Offset [µm]	0.21
Y-Coordinate Pin Offset [µm]	0.00
Supply Rail Width [µm]	0.32
Supply Rail Metal Layer	1
Standard Cell Height [µm]	3.36
Standard Cell Height [Tracks]	8

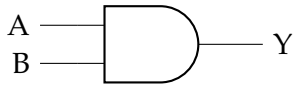
Characterization Parameters

Process Corner	TT
Supply [V]	1.20
Temperature [°C]	25
Input Transition Time [ns]	0.64
Capacitive Load at Each Output [pF]	0.02×i for Xi

AND2

Function

2-input AND.



$$Y = A \cdot B$$

Dimensions

Cell	Height [µm]	Width [µm]
AND2X1	3.36	2.10
AND2X1P5	3.36	2.10
AND2X2	3.36	2.10
AND2X3	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
AND2X1	0.0016	0.0016
AND2X1P5	0.0016	0.0016
AND2X2	0.0016	0.0016
AND2X3	0.0016	0.0016

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
AND2X1	0.3572	0.3651	0.3049	0.3257
AND2X1P5	0.3838	0.3889	0.3239	0.3435
AND2X2	0.4072	0.4089	0.3468	0.3631
AND2X3	0.4529	0.4518	0.3833	0.3982

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	B→Y
AND2X1	0.00370	0.00395
AND2X1P5	0.00470	0.00495
AND2X2	0.00580	0.00605
AND2X3	0.00715	0.00740

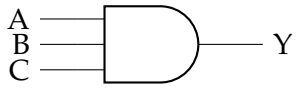
Leakage power

Cell	Leakage [nW]
AND2X1	0.0585
AND2X1P5	0.0750
AND2X2	0.0907
AND2X3	0.1113

AND3

Function

3-input AND.



$$Y = A \cdot B \cdot C$$

Dimensions

Cell	Height [µm]	Width [µm]
AND3X1	3.36	2.52
AND3X2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
AND3X1	0.0016	0.0016	0.0016
AND3X2	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
AND3X1	0.4143	0.4253	0.4225	0.3248	0.3439	0.3573
AND3X2	0.4677	0.4741	0.4666	0.3619	0.3776	0.3894

Internal dynamic energy per switching event

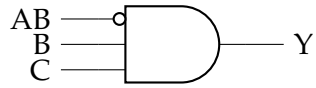
Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
AND3X1	0.00400	0.00430	0.00465
AND3X2	0.00605	0.00640	0.00670

Leakage power

Cell	Leakage [nW]
AND3X1	0.0620
AND3X2	0.0961

AND3B

Function



3-input AND with one inverted input.

$$Y = \overline{AB} \cdot B \cdot C$$

Dimensions

Cell	Height [µm]	Width [µm]
AND3BX1	3.36	2.52
AND3BX2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]		
	AB	B	C
AND3BX1	0.0016	0.0016	0.0016
AND3BX2	0.0033	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	AB→Y	B→Y	C→Y	AB→Y	B→Y	C→Y
AND3BX1	0.5331	0.5153	0.5206	0.3900	0.3167	0.3361
AND3BX2	0.5253	0.5556	0.5563	0.3784	0.3507	0.3665

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	AB→Y	B→Y	C→Y
AND3BX1	0.00130	0.00450	0.00480
AND3BX2	0.00265	0.00715	0.00740

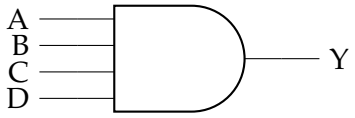
Leakage power

Cell	Leakage [nW]
AND3BX1	0.0572
AND3BX2	0.0927

AND4

Function

4-input AND.



$$Y = A \cdot B \cdot C \cdot D$$

Dimensions

Cell	Height [µm]	Width [µm]
AND4X1	3.36	2.94
AND4X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AND4X1	0.0016	0.0016	0.0016	0.0016
AND4X2	0.0016	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AND4X1	0.4788	0.4934	0.4947	0.4908	0.3459	0.3631	0.3765	0.3865
AND4X2	0.5363	0.5472	0.5448	0.5391	0.3797	0.3928	0.4049	0.4146

Internal dynamic energy per switching event

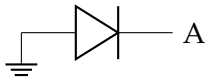
Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AND4X1	0.00440	0.00470	0.00505	0.00540
AND4X2	0.00655	0.00685	0.00720	0.00755

Leakage power

Cell	Leakage [nW]
AND4X1	0.0638
AND4X2	0.0989

ANTENNA

Function



Cell to prevent damage due to process antenna effect.

Dimensions

Cell	Height [μm]	Width [μm]
ANTENNA2	3.36	0.84
ANTENNA4	3.36	1.68
ANTENNA8	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]
	A
ANTENNA2	0.0011
ANTENNA4	0.0023
ANTENNA8	0.0063

Leakage power

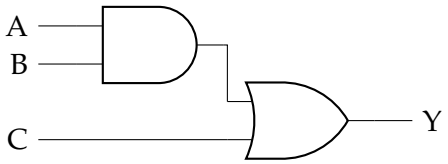
Cell	Leakage [nW]
ANTENNA2	0.0006
ANTENNA4	0.0006
ANTENNA8	0.0006

AO21

Function

A 2-input AND followed by a 2-input OR.

$$Y = A \cdot B + C$$



Dimensions

Cell	Height [μm]	Width [μm]
AO21X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
AO21X2	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
AO21X2	0.4199	0.4183	0.4130	0.4449	0.4222	0.4469

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
AO21X2	0.00625	0.00600	0.00715

Leakage power

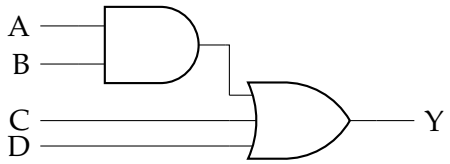
Cell	Leakage [nW]
AO21X2	0.0853

AO211

Function

A 2-input AND followed by a 3-input OR.

$$Y = A \cdot B + C + D$$



Dimensions

Cell	Height [µm]	Width [µm]
AO211X1	3.36	2.94
AO211X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AO211X1	0.0016	0.0016	0.0016	0.0016
AO211X2	0.0016	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AO211X1	0.3788	0.3838	0.3856	0.3980	0.4279	0.4569	0.4786	0.4675
AO211X2	0.4230	0.4232	0.4133	0.4253	0.4888	0.5138	0.5299	0.5148

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AO211X1	0.00400	0.00425	0.00500	0.00555
AO211X2	0.00605	0.00630	0.00715	0.00765

Leakage power

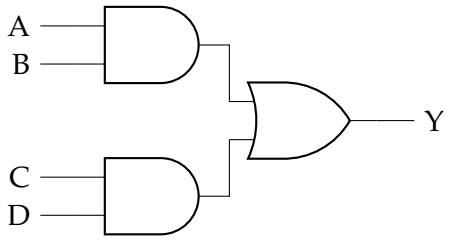
Cell	Leakage [nW]
AO211X1	0.0565
AO211X2	0.0798

AO22

Function

Two 2-input ANDs followed by a 2-input OR.

$$Y = A \cdot B + C \cdot D$$



Dimensions

Cell	Height [µm]	Width [µm]
AO22X1	3.36	3.78
AO22X2	3.36	3.78

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AO22X1	0.0015	0.0016	0.0016	0.0016
AO22X2	0.0015	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AO22X1	0.4430	0.4433	0.4026	0.4002	0.4416	0.4213	0.4240	0.4005
AO22X2	0.4751	0.4796	0.4392	0.4401	0.4770	0.4589	0.4678	0.4473

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AO22X1	0.00565	0.00535	0.00460	0.00435
AO22X2	0.00770	0.00740	0.00670	0.00640

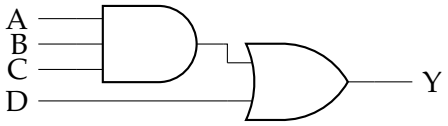
Leakage power

Cell	Leakage [nW]
AO22X1	0.0742
AO22X2	0.1034

AO31

Function

A 3-input AND followed by a 2-input OR.



$$Y = A \cdot B \cdot C + D$$

Dimensions

Cell	Height [μm]	Width [μm]
AO31X1	3.36	3.36
AO31X2	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AO31X1	0.0016	0.0015	0.0015	0.0016
AO31X2	0.0016	0.0015	0.0015	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AO31X1	0.4400	0.4519	0.4460	0.3591	0.3776	0.3996	0.4176	0.4019
AO31X2	0.4904	0.4980	0.4879	0.3900	0.4201	0.4399	0.4561	0.4482

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AO31X1	0.00445	0.00475	0.00515	0.00440
AO31X2	0.00655	0.00685	0.00720	0.00650

Leakage power

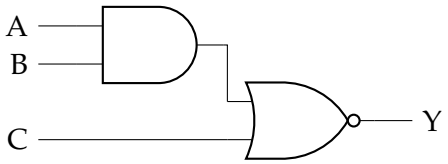
Cell	Leakage [nW]
AO31X1	0.0699
AO31X2	0.0971

AOI21

Function

A 2-input AND followed by a 2-input NOR.

$$Y = \overline{A \cdot B + C}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI21X1	3.36	2.10
AOI21X2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
AOI21X1	0.0015	0.0016	0.0016
AOI21X2	0.0031	0.0033	0.0033

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
AOI21X1	0.5332	0.5222	0.5448	0.4918	0.4996	0.3907
AOI21X2	0.5217	0.5108	0.5344	0.4658	0.4799	0.3796

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
AOI21X1	0.00250	0.00215	0.00180
AOI21X2	0.00490	0.00420	0.00345

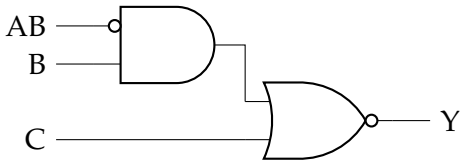
Leakage power

Cell	Leakage [nW]
AOI21X1	0.0410
AOI21X2	0.0946

AOI2B1

Function

A 2-input AND with one inverted input followed by a 2-input NOR.



$$Y = \overline{\overline{AB} \cdot B + C}$$

Dimensions

Cell	Height [µm]	Width [µm]
AOI2B1X1	3.36	2.52
AOI2B1X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	AB	B	C
AOI2B1X1	0.0016	0.0016	0.0016
AOI2B1X2	0.0016	0.0033	0.0033

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	AB→Y	B→Y	C→Y	AB→Y	B→Y	C→Y
AOI2B1X1	0.4552	0.5460	0.5330	0.3934	0.4773	0.4057
AOI2B1X2	0.4945	0.5398	0.5249	0.4253	0.4563	0.3944

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	AB→Y	B→Y	C→Y
AOI2B1X1	0.00395	0.00180	0.00295
AOI2B1X2	0.00660	0.00385	0.00615

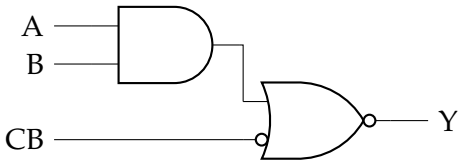
Leakage power

Cell	Leakage [nW]
AOI2B1X1	0.0615
AOI2B1X2	0.1109

AOI21B

Function

A 2-input AND followed by a 2-input NOR whose other input is inverted.



$$Y = \overline{A \cdot B + \overline{CB}}$$

Dimensions

Cell	Height [µm]	Width [µm]
AOI21BX1	3.36	2.52
AOI21BX2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	CB
AOI21BX1	0.0016	0.0016	0.0016
AOI21BX2	0.0033	0.0033	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	CB→Y	A→Y	B→Y	CB→Y
AOI21BX1	0.5348	0.5471	0.4868	0.4870	0.4771	0.3037
AOI21BX2	0.5284	0.5400	0.5274	0.4689	0.4544	0.3467

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	CB→Y
AOI21BX1	0.00145	0.00185	0.00515
AOI21BX2	0.00310	0.00380	0.00890

Leakage power

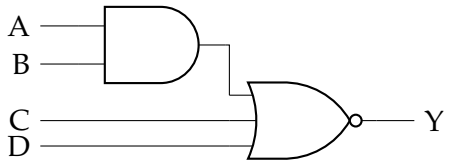
Cell	Leakage [nW]
AOI21BX1	0.0615
AOI21BX2	0.1109

AOI211

Function

A 2-input AND followed by a 3-input NOR.

$$Y = \overline{A \cdot B + C + D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI211X1	3.36	2.52
AOI211X2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AOI211X1	0.0015	0.0016	0.0016	0.0016
AOI211X2	0.0031	0.0032	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AOI211X1	0.6764	0.6599	0.6836	0.6835	0.5043	0.5123	0.4024	0.3944
AOI211X2	0.6601	0.6451	0.6725	0.6745	0.4770	0.4912	0.3905	0.3834

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AOI211X1	0.00340	0.00305	0.00255	0.00205
AOI211X2	0.00640	0.00580	0.00490	0.00395

Leakage power

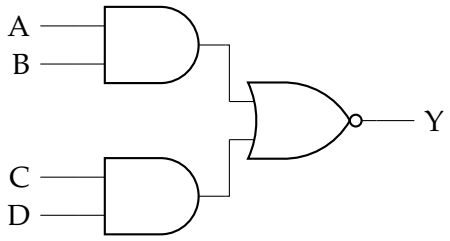
Cell	Leakage [nW]
AOI211X1	0.0435
AOI211X2	0.1040

AOI22

Function

Two 2-input ANDs followed by a 2-input NOR.

$$Y = \overline{A \cdot B + C \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI22X1	3.36	2.94
AOI22X1P5	3.36	2.94
AOI22X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AOI22X1	0.0015	0.0016	0.0016	0.0016
AOI22X1P5	0.0023	0.0024	0.0025	0.0024
AOI22X2	0.0031	0.0032	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AOI22X1	0.5579	0.5465	0.5615	0.5507	0.5060	0.5138	0.4819	0.4902
AOI22X1P5	0.5523	0.5413	0.5470	0.5576	0.4866	0.4990	0.4762	0.4640
AOI22X2	0.5437	0.5330	0.5502	0.5402	0.4784	0.4925	0.4556	0.4699

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AOI22X1	0.00355	0.00315	0.00250	0.00210
AOI22X1P5	0.00510	0.00455	0.00300	0.00350
AOI22X2	0.00670	0.00605	0.00460	0.00395

Leakage power

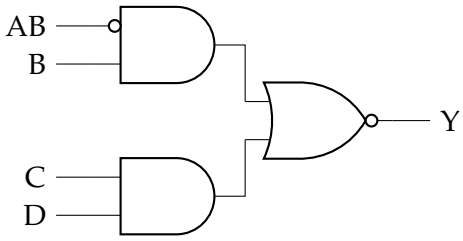
Cell	Leakage [nW]
AOI22X1	0.0491
AOI22X1P5	0.0816
AOI22X2	0.1136

AOI2B2

Function

Two 2-input ANDs, with one of them having one input inverted, followed by a 2-input NOR.

$$Y = \overline{\overline{AB} \cdot B + C \cdot D}$$



Dimensions

Cell	Height [μm]	Width [μm]
AOI2B2X1	3.36	3.36
AOI2B2X2	3.36	3.78

Input pin capacitance

Cell	Capacitance [pF]			
	AB	B	C	D
AOI2B2X1	0.0016	0.0016	0.0016	0.0016
AOI2B2X2	0.0016	0.0033	0.0033	0.0031

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	B→Y	C→Y	D→Y	AB→Y	B→Y	C→Y	D→Y
AOI2B2X1	0.5022	0.5420	0.5568	0.5471	0.4285	0.5112	0.4775	0.4871
AOI2B2X2	0.5074	0.5495	0.5324	0.5427	0.4293	0.4579	0.4925	0.4782

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	B→Y	C→Y	D→Y
AOI2B2X1	0.00585	0.00295	0.00225	0.00190
AOI2B2X2	0.00745	0.00465	0.00675	0.00745

Leakage power

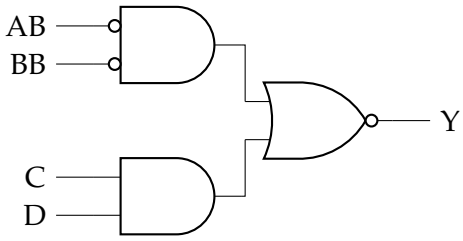
Cell	Leakage [nW]
AOI2B2X1	0.0730
AOI2B2X2	0.1375

AOI2BB2

Function

Two 2-input ANDs, one of them with negated inputs, followed by a 2-input NOR.

$$Y = \overline{\overline{AB} \cdot \overline{BB} + C \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI2BB2X1	3.36	3.78
AOI2BB2X2	3.36	3.78

Input pin capacitance

Cell	Capacitance [pF]			
	AB	BB	C	D
AOI2BB2X1	0.0016	0.0016	0.0015	0.0016
AOI2BB2X2	0.0016	0.0016	0.0031	0.0032

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	BB→Y	C→Y	D→Y	AB→Y	BB→Y	C→Y	D→Y
AOI2BB2X1	0.5015	0.4773	0.5354	0.5235	0.3591	0.3564	0.4929	0.5008
AOI2BB2X2	0.5293	0.5088	0.5227	0.5119	0.4036	0.4090	0.4659	0.4801

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	BB→Y	C→Y	D→Y
AOI2BB2X1	0.00495	0.00450	0.00260	0.00225
AOI2BB2X2	0.00760	0.00710	0.00490	0.00425

Leakage power

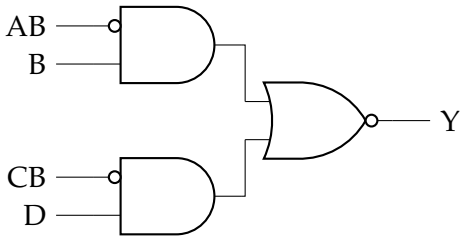
Cell	Leakage [nW]
AOI2BB2X1	0.0694
AOI2BB2X2	0.1213

AOI2B2B

Function

Two 2-input ANDs with 1 inverted input each followed by a 2-input NOR.

$$Y = \overline{\overline{AB} \cdot B + \overline{CB} \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI2B2BX1	3.36	4.20
AOI2B2BX2	3.36	4.20

Input pin capacitance

Cell	Capacitance [pF]			
	AB	B	CB	D
AOI2B2BX1	0.0016	0.0016	0.0016	0.0015
AOI2B2BX2	0.0016	0.0033	0.0016	0.0031

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	B→Y	CB→Y	D→Y	AB→Y	B→Y	CB→Y	D→Y
AOI2B2BX1	0.4695	0.5597	0.5002	0.5591	0.3961	0.4816	0.4215	0.5078
AOI2B2BX2	0.5072	0.5500	0.5352	0.5431	0.4267	0.4571	0.4457	0.4796

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	B→Y	CB→Y	D→Y
AOI2B2BX1	0.00450	0.00235	0.00600	0.00390
AOI2B2BX2	0.00745	0.00450	0.01020	0.00735

Leakage power

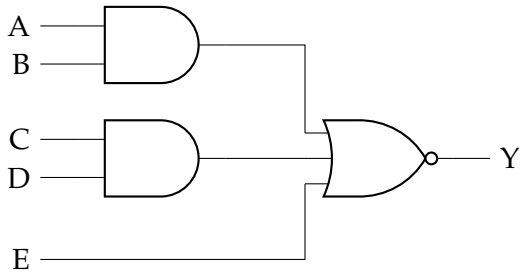
Cell	Leakage [nW]
AOI2B2BX1	0.0969
AOI2B2BX2	0.1614

AOI221

Function

Two 2-input ANDs followed by a 3-input NOR.

$$Y = \overline{A \cdot B + C \cdot D + E}$$



Dimensions

Cell	Height [μm]	Width [μm]
AOI221X1	3.36	2.94
AOI221X2	3.36	3.36
AOI221X3	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]				
	A	B	C	D	E
AOI221X1	0.0016	0.0015	0.0016	0.0016	0.0016
AOI221X2	0.0032	0.0031	0.0033	0.0033	0.0033
AOI221X3	0.0044	0.0042	0.0044	0.0044	0.0044

Delay

Cell	Rising delay [ns]				
	A→Y	B→Y	C→Y	D→Y	E→Y
AOI221X1	0.7002	0.7156	0.6889	0.7054	0.7239
AOI221X2	0.6777	0.6927	0.6776	0.6939	0.7050
AOI221X3	0.7227	0.7370	0.7205	0.7395	0.7485

Cell	Falling delay [ns]				
	A→Y	B→Y	C→Y	D→Y	E→Y
AOI221X1	0.5271	0.5209	0.4944	0.4863	0.4135
AOI221X2	0.5035	0.4898	0.4742	0.4597	0.3986
AOI221X3	0.5181	0.5030	0.4900	0.4741	0.4107

Internal dynamic energy per switching event

Cell	Energy [pJ]				
	A→Y	B→Y	C→Y	D→Y	E→Y
AOI221X1	0.00410	0.00450	0.00235	0.00270	0.00370
AOI221X2	0.00760	0.00825	0.00440	0.00505	0.00680
AOI221X3	0.01000	0.01090	0.00570	0.00675	0.00890

Leakage power

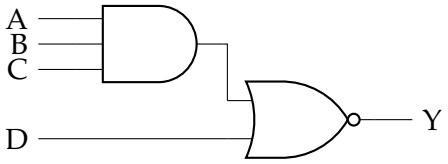
Cell	Leakage [nW]
AOI221X1	0.0516
AOI221X2	0.1234
AOI221X3	0.1712

AOI31

Function

A 3-input AND followed by a 2-input NOR.

$$Y = \overline{A \cdot B \cdot C + D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOI31X1	3.36	2.52
AOI31X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AOI31X1	0.0016	0.0015	0.0015	0.0016
AOI31X2	0.0033	0.0031	0.0032	0.0034

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AOI31X1	0.5275	0.5385	0.5477	0.5550	0.5941	0.6010	0.5840	0.3909
AOI31X2	0.5169	0.5273	0.5354	0.5459	0.5688	0.5690	0.5481	0.3800

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AOI31X1	0.00235	0.00275	0.00315	0.00225
AOI31X2	0.00465	0.00535	0.00605	0.00445

Leakage power

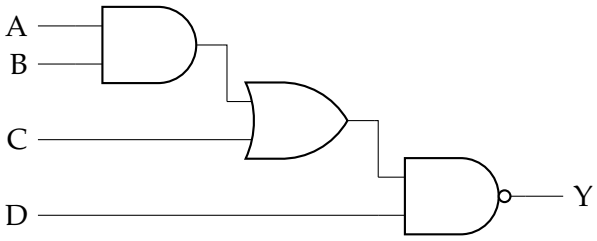
Cell	Leakage [nW]
AOI31X1	0.0472
AOI31X2	0.1105

AOAI211

Function

A 2-input AND followed by a 2-input OR followed by a 2-input NAND.

$$Y = \overline{(A \cdot B + C) \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
AOAI211X1	3.36	2.94
AOAI211X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
AOAI211X1	0.0015	0.0015	0.0016	0.0016
AOAI211X2	0.0031	0.0031	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
AOAI211X1	0.5497	0.5404	0.5618	0.4095	0.5894	0.6058	0.4933	0.5918
AOAI211X2	0.5379	0.5297	0.5518	0.3987	0.5531	0.5735	0.4674	0.5672

Internal dynamic energy per switching event

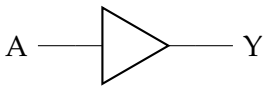
Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
AOAI211X1	0.00320	0.00280	0.00250	0.00220
AOAI211X2	0.00625	0.00560	0.00470	0.00435

Leakage power

Cell	Leakage [nW]
AOAI211X1	0.0476
AOAI211X2	0.1116

BUF

Function



Buffers the input at the output.

$$Y = A$$

Dimensions

Cell	Height [µm]	Width [µm]
BUFX1	3.36	1.68
BUFX2	3.36	1.68
BUFX3	3.36	1.68
BUFX4	3.36	2.10
BUFX6	3.36	2.10
BUFX8	3.36	3.36
BUFX12	3.36	3.36
BUFX16	3.36	4.62
BUFX24	3.36	6.72
BUFX32	3.36	8.82
BUFX44	3.36	10.92
BUFX48	3.36	12.60
BUFX64	3.36	15.96

Input pin capacitance

Cell	Capacitance [pF]
	A
BUFX1	0.0016
BUFX2	0.0016
BUFX3	0.0016
BUFX4	0.0016
BUFX6	0.0025
BUFX8	0.0033
BUFX12	0.0045
BUFX16	0.0066
BUFX24	0.0134
BUFX32	0.0133
BUFX44	0.0179
BUFX48	0.0224
BUFX64	0.0268

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	A→Y	A→Y
BUFX1	0.3138	0.2940
BUFX2	0.3534	0.3356
BUFX3	0.3920	0.3718
BUFX4	0.4181	0.4091
BUFX6	0.4123	0.3993
BUFX8	0.4057	0.3992
BUFX12	0.4151	0.4078
BUFX16	0.4023	0.3928
BUFX24	0.3690	0.3592
BUFX32	0.4038	0.3942
BUFX44	0.4022	0.3962
BUFX48	0.3881	0.3814
BUFX64	0.4005	0.3950

Internal dynamic energy per switching event

Cell	Energy [pJ]
	A→Y
BUFX1	0.00355
BUFX2	0.00560
BUFX3	0.00700
BUFX4	0.00925
BUFX6	0.01270
BUFX8	0.01880
BUFX12	0.02535
BUFX16	0.03730
BUFX24	0.05975
BUFX32	0.07475
BUFX44	0.10015
BUFX48	0.11515
BUFX64	0.15045

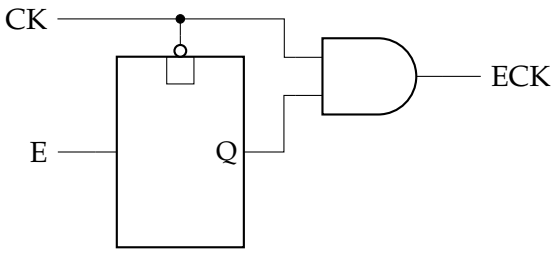
Leakage power

Cell	Leakage [nW]
BUFX1	0.0478
BUFX2	0.0760
BUFX3	0.0946
BUFX4	0.1282
BUFX6	0.1796
BUFX8	0.2606
BUFX12	0.3535
BUFX16	0.5284
BUFX24	0.8484
BUFX32	1.0569
BUFX44	1.4140
BUFX48	1.6261
BUFX64	2.1210

CG

Function

Clock gate for positive-edge-triggered flip-flops. An integrated negative latch samples the enable input E while CK is low; the latched value gates the clock.



$$\mathrm{ECK} = \begin{cases} 0 & \text{if CK} = 0 \\ \mathrm{Q} \cdot \mathrm{CK} & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
CGX1	3.36	6.30
CGX2	3.36	6.30
CGX3	3.36	6.30

Input pin capacitance

Cell	Capacitance [pF]	
	CK	E
CGX1	0.0016	0.0015
CGX2	0.0016	0.0015
CGX3	0.0016	0.0015

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	CK→ECK	CK→ECK
CGX1	0.4710	0.4344
CGX2	0.4940	0.4445
CGX3	0.5258	0.4635

Timing constraints

Cell	Setup time [ns]	Hold time [ns]	Min. pulse [ns]
	E↤CK	CK↦E	CK
CGX1	0.2024	0.0938	3.5156
CGX2	0.2024	0.0938	3.5156
CGX3	0.2024	0.0938	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]
	CK→ECK
CGX1	0.01530
CGX2	0.01755
CGX3	0.01900

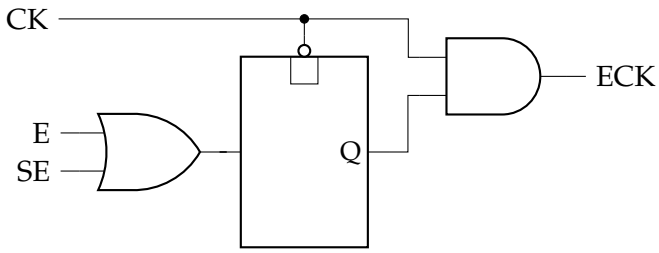
Leakage power

Cell	Leakage [nW]
CGX1	0.2034
CGX2	0.2343
CGX3	0.2542

CGTS

Function

Clock gate with test/scan enable for positive-edge-triggered flip-flops. An integrated negative latch samples E+SE while CK is low; the latched value gates the clock.



$$\mathrm{ECK} = \begin{cases} 0 & \text{if CK} = 0 \\ \mathrm{Q} \cdot \mathrm{CK} & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
CGTSX1	3.36	7.56
CGTSX1P5	3.36	7.56
CGTSX2	3.36	7.56
CGTSX3	3.36	7.56

Input pin capacitance

Cell	Capacitance [pF]		
	CK	E	SE
CGTSX1	0.0016	0.0016	0.0016
CGTSX1P5	0.0016	0.0016	0.0016
CGTSX2	0.0016	0.0016	0.0016
CGTSX3	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	CK→ECK	CK→ECK
CGTSX1	0.4722	0.4358
CGTSX1P5	0.4853	0.4391
CGTSX2	0.4952	0.4454
CGTSX3	0.5270	0.4650

Timing constraints

Cell	Setup time [ns]		Hold time [ns]		Min. pulse [ns]
	E↤CK	SE↤CK	CK↦E	CK↦SE	CK
CGTSX1	0.2641	0.2542	0.0247	-0.0247	3.5156
CGTSX1P5	0.2641	0.2542	0.0247	-0.0247	3.5156
CGTSX2	0.2641	0.2542	0.0247	-0.0247	3.5156
CGTSX3	0.2641	0.2542	0.0247	-0.0247	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]
	CK→ECK
CGTSX1	0.01530
CGTSX1P5	0.01640
CGTSX2	0.01750
CGTSX3	0.01900

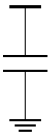
Leakage power

Cell	Leakage [nW]
CGTSX1	0.2353
CGTSX1P5	0.2510
CGTSX2	0.2662
CGTSX3	0.2861

DECAP

Function

Filler cells with decoupling capacitance between power rails.



Dimensions

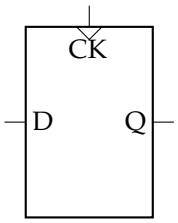
Cell	Height [µm]	Width [µm]
DECAP4	3.36	1.68
DECAP8	3.36	3.36
DECAP16	3.36	6.72
DECAP32	3.36	13.44

Leakage power

Cell	Leakage [nW]
DECAP4	0.3931
DECAP8	1.2557
DECAP16	3.1797
DECAP32	6.6637

DFFQ

Function



Positive edge-triggered D-type flip-flop with a single non-inverted output Q.

$$Q = \begin{cases} D & \text{if CK} \uparrow \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
DFFQX1	3.36	6.72
DFFQX1P5	3.36	7.14
DFFQX2	3.36	7.14
DFFQX3	3.36	7.14
DFFQX4	3.36	7.98
DFFQX6	3.36	7.98

Input pin capacitance

Cell	Capacitance [pF]	
	CK	D
DFFQX1	0.0016	0.0016
DFFQX1P5	0.0016	0.0016
DFFQX2	0.0016	0.0016
DFFQX3	0.0016	0.0016
DFFQX4	0.0016	0.0016
DFFQX6	0.0016	0.0016

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	CK→Q	CK→Q
DFFQX1	0.5171	0.5051
DFFQX1P5	0.5250	0.5087
DFFQX2	0.5303	0.5157
DFFQX3	0.5555	0.5365
DFFQX4	0.5579	0.5492
DFFQX6	0.5915	0.5810

Timing constraints

Cell	Setup time [ns]	Hold time [ns]	Min. pulse [ns]
	D↤CK	CK↦D	CK
DFFQX1	0.0938	0.1384	3.5156
DFFQX1P5	0.0938	0.1342	3.5156
DFFQX2	0.0938	0.1360	3.5156
DFFQX3	0.0938	0.1316	3.5156
DFFQX4	0.0938	0.1329	3.5156
DFFQX6	0.0938	0.1302	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]
	CK→Q
DFFQX1	0.01395
DFFQX1P5	0.01505
DFFQX2	0.01615
DFFQX3	0.01760
DFFQX4	0.02010
DFFQX6	0.02280

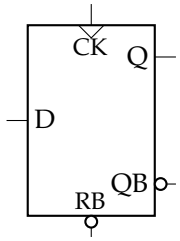
Leakage power

Cell	Leakage [nW]
DFFQX1	0.2252
DFFQX1P5	0.2395
DFFQX2	0.2535
DFFQX3	0.2720
DFFQX4	0.3056
DFFQX6	0.3427

DFFR

Function

Positive edge-triggered D-type flip-flop with asynchronous active-low reset RB and complementary outputs Q and QB.



$$Q = \begin{cases} 0 & \text{if RB} = 0 \\ D & \text{if RB} = 1 \text{ and CK} \uparrow \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
DFFRX1	3.36	9.24
DFFRX1P5	3.36	9.24
DFFRX2	3.36	9.24
DFFRX3	3.36	9.24
DFFRX4	3.36	10.50
DFFRX6	3.36	10.50

Input pin capacitance

Cell	Capacitance [pF]		
	CK	D	RB
DFFRX1	0.0016	0.0016	0.0035
DFFRX1P5	0.0016	0.0016	0.0035
DFFRX2	0.0016	0.0016	0.0035
DFFRX3	0.0016	0.0016	0.0035
DFFRX4	0.0016	0.0016	0.0035
DFFRX6	0.0016	0.0016	0.0035

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	CK→Q	CK→QB	RB→QB	CK→Q	CK→QB	RB→Q
DFFRX1	0.5939	0.5071	0.5269	0.5348	0.5124	0.3755
DFFRX1P5	0.6210	0.5177	0.5385	0.5486	0.5221	0.3890
DFFRX2	0.6446	0.5250	0.5470	0.5651	0.5334	0.4039
DFFRX3	0.6967	0.5556	0.6005	0.6003	0.5630	0.4353
DFFRX4	0.7479	0.5653	0.6493	0.6402	0.5864	0.4622
DFFRX6	0.8162	0.6413	0.7578	0.6853	0.6115	0.5052

Timing constraints

Cell	Setup/rec. time [ns]		Hold/rem. time [ns]		Min. pulse [ns]	
	D↤CK	RB↤CK	CK↦D	CK↦RB	CK	RB
DFFRX1	0.1800	0.1895	0.1027	-0.1752	3.5156	3.5156
DFFRX1P5	0.1802	0.1856	0.1046	-0.1752	3.5156	3.5156
DFFRX2	0.1827	0.1851	0.1053	-0.1752	3.5156	3.5156
DFFRX3	0.1827	0.1851	0.1086	-0.1777	3.5156	3.5156
DFFRX4	0.1851	0.1876	0.1185	-0.1802	3.5156	3.5156
DFFRX6	0.1851	0.1876	0.1185	-0.1802	3.5156	3.5156

Internal dynamic energy per switching event

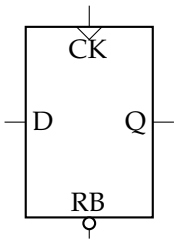
Cell	Energy [pJ]			
	CK→Q	CK→QB	RB→QB	RB→Q
DFFRX1	0.00830	0.00830	0.00990	0.00990
DFFRX1P5	0.00935	0.00935	0.01100	0.01100
DFFRX2	0.01040	0.01040	0.01210	0.01200
DFFRX3	0.01190	0.01190	0.01310	0.01340
DFFRX4	0.01420	0.01420	0.01550	0.01550
DFFRX6	0.01610	0.01610	0.01720	0.01730

Leakage power

Cell	Leakage [nW]
DFFRX1	0.2756
DFFRX1P5	0.3041
DFFRX2	0.3320
DFFRX3	0.3692
DFFRX4	0.4363
DFFRX6	0.4926

DFFRQ

Function



Positive edge-triggered D-type flip-flop with asynchronous active-low reset RB and a single non-inverted output Q.

$$Q = \begin{cases} 0 & \text{if RB} = 0 \\ D & \text{if RB} = 1 \text{ and CK} \uparrow \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [μm]	Width [μm]
DFFRQX1	3.36	8.40
DFFRQX1P5	3.36	8.40
DFFRQX2	3.36	8.40
DFFRQX3	3.36	8.40
DFFRQX4	3.36	9.24
DFFRQX6	3.36	9.24

Input pin capacitance

Cell	Capacitance [pF]		
	CK	D	RB
DFFRQX1	0.0016	0.0016	0.0037
DFFRQX1P5	0.0016	0.0016	0.0037
DFFRQX2	0.0016	0.0016	0.0037
DFFRQX3	0.0016	0.0016	0.0037
DFFRQX4	0.0016	0.0016	0.0037
DFFRQX6	0.0016	0.0016	0.0037

Delay

Cell	Rising delay [ns]	Falling delay [ns]	
	CK→Q	CK→Q	RB→Q
DFFRQX1	0.5699	0.5219	0.3523
DFFRQX1P5	0.5831	0.5257	0.3683
DFFRQX2	0.5929	0.5323	0.3848
DFFRQX3	0.6253	0.5534	0.4197
DFFRQX4	0.6396	0.5647	0.4469
DFFRQX6	0.6875	0.5968	0.4909

Timing constraints

Cell	Setup/rec. time [ns]		Hold/rem. time [ns]		Min. pulse [ns]	
	D⇠⇢CK	RB⇠⇢CK	CK⇢↦D	CK⇢↦RB	CK	RB
DFFRQX1	0.1797	0.2325	0.1094	-0.1728	3.5156	3.5156
DFFRQX1P5	0.1797	0.2457	0.1089	-0.1728	3.5156	3.5156
DFFRQX2	0.1797	0.2592	0.1116	-0.1728	3.5156	3.5156
DFFRQX3	0.1797	0.2819	0.1105	-0.1728	3.5156	3.5156
DFFRQX4	0.1797	0.3302	0.1062	-0.1728	3.5156	3.5156
DFFRQX6	0.1777	0.3958	0.1066	-0.1728	3.5156	3.5156

Internal dynamic energy per switching event

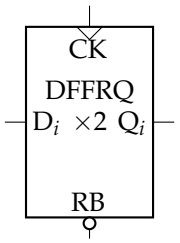
Cell	Energy [pJ]	
	CK→Q	RB→Q
DFFRQX1	0.01490	0.01710
DFFRQX1P5	0.01600	0.01820
DFFRQX2	0.01710	0.01910
DFFRQX3	0.01860	0.01990
DFFRQX4	0.02085	0.02200
DFFRQX6	0.02360	0.02450

Leakage power

Cell	Leakage [nW]
DFFRQX1	0.2526
DFFRQX1P5	0.2683
DFFRQX2	0.2834
DFFRQX3	0.3034
DFFRQX4	0.3415
DFFRQX6	0.3813

DFFRQ2

Function



Bundled pair of positive edge-triggered D-type flip-flops sharing clock circuitry and asynchronous active-low reset RB signal, with non-inverted outputs Q0 and Q1.

$$Q_i = \begin{cases} 0 & \text{if RB} = 0 \\ D_i & \text{if RB} = 1 \text{ and CK} \uparrow \\ Q_i & \text{otherwise} \end{cases} \quad (i \in \{0,1\})$$

Dimensions

Cell	Height [µm]	Width [µm]
DFFRQ2X1	3.36	14.70
DFFRQ2X1P5	3.36	14.70
DFFRQ2X2	3.36	14.70
DFFRQ2X3	3.36	14.70

Input pin capacitance

Cell	Capacitance [pF]			
	CK	D0	D1	RB
DFFRQ2X1	0.0016	0.0016	0.0016	0.0068
DFFRQ2X1P5	0.0016	0.0016	0.0016	0.0068
DFFRQ2X2	0.0016	0.0016	0.0016	0.0068
DFFRQ2X3	0.0016	0.0016	0.0016	0.0068

Delay

Cell	Rising delay [ns]		Falling delay [ns]			
	CK→Q0	CK→Q1	CK→Q0	CK→Q1	RB→Q0	RB→Q1
DFFRQ2X1	0.6013	0.6012	0.5759	0.5767	0.3518	0.3520
DFFRQ2X1P5	0.6141	0.6141	0.5795	0.5803	0.3675	0.3677
DFFRQ2X2	0.6249	0.6250	0.5868	0.5877	0.3850	0.3852
DFFRQ2X3	0.6567	0.6566	0.6078	0.6087	0.4196	0.4199

Timing constraints

Cell	Setup/rec. time [ns]			Hold/rem. time [ns]			Min. pulse [ns]	
	D0↤CK	D1↤CK	RB↤CK	CK↦D0	CK↦D1	CK↦RB	CK	RB
DFFRQ2X1	0.1252	0.1250	0.1677	0.2370	0.2370	-0.1259	3.5156	3.5156
DFFRQ2X1P5	0.1247	0.1245	0.1744	0.2370	0.2370	-0.1259	3.5156	3.5156
DFFRQ2X2	0.1261	0.1239	0.1830	0.2370	0.2370	-0.1259	3.5156	3.5156
DFFRQ2X3	0.1257	0.1231	0.1926	0.2326	0.2370	-0.1259	3.5156	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	CK→Q0	CK→Q1	RB→Q0	RB→Q1
DFFRQ2X1	0.01800	0.01800	0.02540	0.02540
DFFRQ2X1P5	0.01915	0.01910	0.02640	0.02640
DFFRQ2X2	0.02025	0.02030	0.02730	0.02730
DFFRQ2X3	0.02175	0.02180	0.02880	0.02880

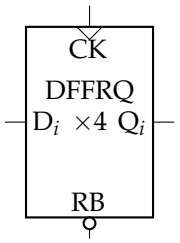
Leakage power

Cell	Leakage [nW]
DFFRQ2X1	0.4292
DFFRQ2X1P5	0.4606
DFFRQ2X2	0.4909
DFFRQ2X3	0.5307

DFFRQ4

Function

Bundled pair of positive edge-triggered D-type flip-flops sharing clock circuitry and asynchronous active-low reset RB signal, with non-inverted outputs Q0 through Q3.



$$Q_i = \begin{cases} 0 & \text{if RB} = 0 \\ D_i & \text{if RB} = 1 \text{ and CK} \uparrow \\ Q_i & \text{otherwise} \end{cases} \quad (i \in \{0,1,2,3\})$$

Dimensions

Cell	Height [µm]	Width [µm]
DFFRQ4X1	3.36	28.56
DFFRQ4X1P5	3.36	28.56
DFFRQ4X2	3.36	28.56
DFFRQ4X3	3.36	28.56

Input pin capacitance

Cell	Capacitance [pF]					
	CK	D0	D1	D2	D3	RB
DFFRQ4X1	0.0025	0.0016	0.0016	0.0016	0.0016	0.0131
DFFRQ4X1P5	0.0025	0.0016	0.0016	0.0016	0.0016	0.0131
DFFRQ4X2	0.0025	0.0016	0.0016	0.0016	0.0016	0.0131
DFFRQ4X3	0.0025	0.0016	0.0016	0.0016	0.0016	0.0131

Delay

Cell	Rising delay [ns]			
	CK→Q0	CK→Q1	CK→Q2	CK→Q3
DFFRQ4X1	0.6021	0.6019	0.6020	0.6019
DFFRQ4X1P5	0.6149	0.6146	0.6144	0.6143
DFFRQ4X2	0.6256	0.6254	0.6249	0.6249
DFFRQ4X3	0.6574	0.6572	0.6564	0.6565

Cell	Falling delay [ns]							
	CK→Q0	CK→Q1	CK→Q2	CK→Q3	RB→Q0	RB→Q1	RB→Q2	RB→Q3
DFFRQ4X1	0.6029	0.6026	0.6028	0.6029	0.3532	0.3534	0.3529	0.3530
DFFRQ4X1P5	0.6063	0.6062	0.6063	0.6064	0.3689	0.3690	0.3682	0.3683
DFFRQ4X2	0.6136	0.6136	0.6135	0.6136	0.3862	0.3864	0.3852	0.3854
DFFRQ4X3	0.6347	0.6348	0.6345	0.6347	0.4206	0.4209	0.4194	0.4197

Timing constraints

Cell	Setup/rec. time [ns]				
	D0↤CK	D1↤CK	D2↤CK	D3↤CK	RB↤CK
DFFRQ4X1	0.1287	0.1288	0.1311	0.1300	0.1704
DFFRQ4X1P5	0.1284	0.1264	0.1306	0.1297	0.1739
DFFRQ4X2	0.1280	0.1280	0.1304	0.1293	0.1809
DFFRQ4X3	0.1270	0.1277	0.1299	0.1290	0.1871

Cell	Hold/rem. time [ns]					Min. pulse [ns]	
	CK↦D0	CK↦D1	CK↦D2	CK↦D3	CK↦RB	CK	RB
DFFRQ4X1	0.2820	0.2824	0.2832	0.2820	-0.1283	3.5156	3.5156
DFFRQ4X1P5	0.2817	0.2821	0.2829	0.2857	-0.1283	3.5156	3.5156
DFFRQ4X2	0.2812	0.2855	0.2824	0.2851	-0.1283	3.5156	3.5156
DFFRQ4X3	0.2841	0.2805	0.2854	0.2841	-0.1283	3.5156	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]							
	CK→Q0	CK→Q1	CK→Q2	CK→Q3	RB→Q0	RB→Q1	RB→Q2	RB→Q3
DFFRQ4X1	0.02350	0.02350	0.02350	0.02350	0.04340	0.04330	0.04340	0.04340
DFFRQ4X1P5	0.02465	0.02460	0.02460	0.02460	0.04440	0.04440	0.04430	0.04430
DFFRQ4X2	0.02575	0.02575	0.02575	0.02575	0.04530	0.04530	0.04530	0.04530
DFFRQ4X3	0.02715	0.02715	0.02710	0.02710	0.04670	0.04670	0.04670	0.04670

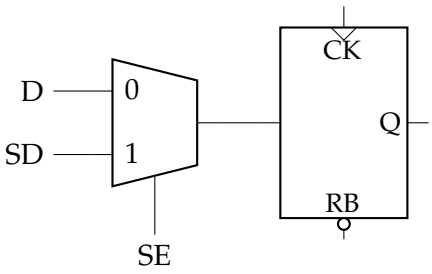
Leakage power

Cell	Leakage [nW]
DFFRQ4X1	0.7966
DFFRQ4X1P5	0.8594
DFFRQ4X2	0.9200
DFFRQ4X3	0.9997

DFFMRQ

Function

Positive edge-triggered D-type flip-flop with asynchronous active-low reset RB, a single non-inverted output Q, and a 2-input multiplexer at its input. Useful, e.g., to implement scan chains.



$$Q = \begin{cases} 0 & \text{if RB} = 0 \\ D & \text{if RB} = 1, \text{CK} \uparrow \text{ and SE} = 0 \\ SD & \text{if RB} = 1, \text{CK} \uparrow \text{ and SE} = 1 \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
DFFMRQX2	3.36	10.92

Input pin capacitance

Cell	Capacitance [pF]				
	CK	D	RB	SD	SE
DFFMRQX2	0.0016	0.0016	0.0038	0.0016	0.0031

Delay

Cell	Rising delay [ns]	Falling delay [ns]	
	CK→Q	CK→Q	RB→Q
DFFMRQX2	0.5965	0.5361	0.3884

Timing constraints

Cell	Setup/rec. time [ns]			
	D↤CK	SD↤CK	SE↤CK	RB↤CK
DFFMRQX2	0.4147	0.4122	0.4173	0.4344

Cell	Hold/rem. time [ns]				Min. pulse [ns]	
	CK↦D	CK↦SD	CK↦SE	CK↦RB	CK	RB
DFFMRQX2	-0.1362	-0.1375	-0.0238	-0.3925	3.5156	3.5156

Internal dynamic energy per switching event

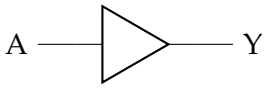
Cell	Energy [pJ]	
	CK→Q	RB→Q
DFFMRQX2	0.01755	0.02310

Leakage power

Cell	Leakage [nW]
DFFMRQX2	0.3384

DLY1

Function



Delay cell (4 inverters in series).

$$Y = A$$

Dimensions

Cell	Height [µm]	Width [µm]
DLY1X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]
	A
DLY1X2	0.0016

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	A→Y	A→Y
DLY1X2	0.3919	0.3686

Internal dynamic energy per switching event

Cell	Energy [pJ]
	A→Y
DLY1X2	0.01000

Leakage power

Cell	Leakage [nW]
DLY1X2	0.1239

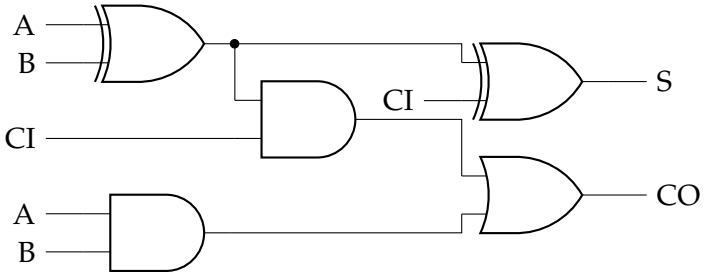
FA

Function

Full adder implementing
$\{CO, S\} = A + B + CI$.

$$S = A \oplus B \oplus CI$$

$$CO = A \cdot B + (A \oplus B) \cdot CI$$



Dimensions

Cell	Height [µm]	Width [µm]
FAX1	3.36	8.40

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	CI
FAX1	0.0015	0.0033	0.0038

Delay

Cell	Rising delay [ns]					
	A→CO	B→CO	CI→CO	A→S	B→S	CI→S
FAX1	0.8359	0.7484	0.3781	0.7323	0.6428	0.5688

Cell	Falling delay [ns]					
	A→CO	B→CO	CI→CO	A→S	B→S	CI→S
FAX1	0.7044	0.6652	0.3981	0.6869	0.6047	0.5065

Internal dynamic energy per switching event

Cell	Energy [pJ]					
	A→CO	B→CO	CI→CO	A→S	B→S	CI→S
FAX1	0.00920	0.00920	0.00485	0.02435	0.01260	0.00880

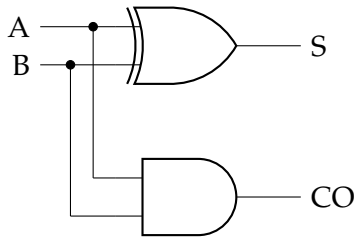
Leakage power

Cell	Leakage [nW]
FAX1	0.2934

HA

Function

Half adder implementing $\{CO, S\} = A + B$.



$$S = A \oplus B$$

$$CO = A \cdot B$$

Dimensions

Cell	Height [µm]	Width [µm]
HAX2	3.36	4.62

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
HAX2	0.0028	0.0026

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→CO	B→CO	A→S	B→S	A→CO	B→CO	A→S	B→S
HAX2	0.4464	0.4518	0.4842	0.4797	0.3975	0.3849	0.4638	0.4674

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→CO	B→CO	A→S	B→S
HAX2	0.00620	0.00625	0.00665	0.00625

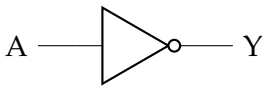
Leakage power

Cell	Leakage [nW]
HAX2	0.1908

INV

Function

Inverter.



$$Y = \overline{A}$$

Dimensions

Cell	Height [µm]	Width [µm]
INVX1	3.36	1.26
INVX2	3.36	1.26
INVX3	3.36	1.26
INVX4	3.36	1.68
INVX6	3.36	1.68
INVX8	3.36	2.10
INVX14	3.36	3.36
INVX16	3.36	3.78
INVX32	3.36	6.72
INVX64	3.36	13.02

Input pin capacitance

Cell	Capacitance [pF]
	A
INVX1	0.0016
INVX2	0.0033
INVX3	0.0044
INVX4	0.0066
INVX6	0.0089
INVX8	0.0134
INVX14	0.0222
INVX16	0.0267
INVX32	0.0534
INVX64	0.1068

Delay

Cell	Rising delay [ns]	Falling delay [ns]
	A→Y	A→Y
INVX1	0.4033	0.3881
INVX2	0.3930	0.3762
INVX3	0.4117	0.3881
INVX4	0.3910	0.3745
INVX6	0.4108	0.3873
INVX8	0.3907	0.3687
INVX14	0.4000	0.3782
INVX16	0.3906	0.3698
INVX32	0.3919	0.3708
INVX64	0.4023	0.3818

Internal dynamic energy per switching event

Cell	Energy [pJ]
	A→Y
INVX1	0.00125
INVX2	0.00240
INVX3	0.00315
INVX4	0.00420
INVX6	0.00570
INVX8	0.00865
INVX14	0.01450
INVX16	0.01640
INVX32	0.03380
INVX64	0.06670

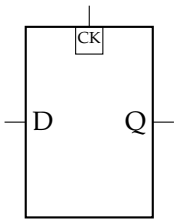
Leakage power

Cell	Leakage [nW]
INVX1	0.0239
INVX2	0.0521
INVX3	0.0707
INVX4	0.1043
INVX6	0.1414
INVX8	0.2121
INVX14	0.3535
INVX16	0.4242
INVX32	0.8484
INVX64	1.6968

LATQ

Function

Positive D-type latch with a single non-inverted output Q.



$$Q = \begin{cases} D & \text{if } CK = 1 \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
LATQX1	3.36	5.46
LATQX2	3.36	5.88
LATQX3	3.36	5.88

Input pin capacitance

Cell	Capacitance [pF]	
	CK	D
LATQX1	0.0016	0.0016
LATQX2	0.0016	0.0016
LATQX3	0.0016	0.0016

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	CK→Q	D→Q	CK→Q	D→Q
LATQX1	0.5067	0.4484	0.4944	0.4160
LATQX2	0.5232	0.4645	0.5085	0.4292
LATQX3	0.5479	0.4890	0.5294	0.4501

Timing constraints

Cell	Setup time [ns]	Hold time [ns]	Min. pulse [ns]
	D↤CK	CK↦D	CK
LATQX1	0.3288	0.0346	3.5156
LATQX2	0.3329	0.0321	3.5156
LATQX3	0.3303	0.0296	3.5156

Internal dynamic energy per switching event

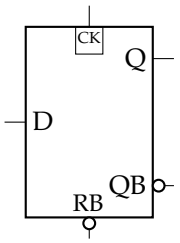
Cell	Energy [pJ]	
	CK→Q	D→Q
LATQX1	0.01265	0.00995
LATQX2	0.01505	0.01225
LATQX3	0.01645	0.01365

Leakage power

Cell	Leakage [nW]
LATQX1	0.1949
LATQX2	0.2231
LATQX3	0.2416

LATR

Function



Positive D-type latch with asynchronous active-low reset RB and complementary outputs Q and QB.

$$Q = \begin{cases} 0 & \text{if RB} = 0 \\ D & \text{if RB} = 1 \text{ and CK} = 1 \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [μm]	Width [μm]
LATRX1	3.36	7.14
LATRX1P5	3.36	7.14
LATRX2	3.36	7.14
LATRX3	3.36	7.14

Input pin capacitance

Cell	Capacitance [pF]		
	CK	D	RB
LATRX1	0.0016	0.0016	0.0021
LATRX1P5	0.0016	0.0016	0.0021
LATRX2	0.0016	0.0016	0.0021
LATRX3	0.0016	0.0016	0.0021

Delay

Cell	Rising delay [ns]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATRX1	0.5438	0.4943	0.4817	0.5590	0.5103	0.4819
LATRX1P5	0.5609	0.5162	0.5020	0.5817	0.5405	0.5045
LATRX2	0.5747	0.5342	0.5187	0.6018	0.5673	0.5239
LATRX3	0.6137	0.5776	0.5609	0.6465	0.6195	0.5665

Cell	Falling delay [ns]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATRX1	0.5148	0.4736	0.4337	0.5616	0.5057	0.4935
LATRX1P5	0.5262	0.4915	0.4442	0.5819	0.5321	0.5185
LATRX2	0.5399	0.5108	0.4562	0.6052	0.5605	0.5455
LATRX3	0.5714	0.5484	0.4836	0.6473	0.6083	0.5916

Timing constraints

Cell	Setup/rec. time [ns]		Hold/rem. time [ns]		Min. pulse [ns]	
	D↤CK	RB↤CK	CK↦D	CK↦RB	CK	RB
LATRX1	0.2592	0.1481	-0.0666	-0.0592	3.5156	3.5156
LATRX1P5	0.3135	0.1950	-0.0938	-0.0815	3.5156	3.5156
LATRX2	0.3628	0.2394	-0.1209	-0.1037	3.5156	3.5156
LATRX3	0.4270	0.2937	-0.1555	-0.1333	3.5156	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATRX1	0.00800	0.00485	0.00535	0.00800	0.00485	0.00540
LATRX1P5	0.00910	0.00595	0.00645	0.00910	0.00595	0.00645
LATRX2	0.01010	0.00695	0.00750	0.01010	0.00695	0.00750
LATRX3	0.01160	0.00835	0.00890	0.01160	0.00840	0.00895

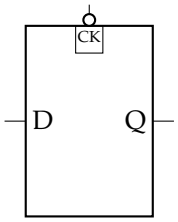
Leakage power

Cell	Leakage [nW]
LATRX1	0.2085
LATRX1P5	0.2371
LATRX2	0.2649
LATRX3	0.3021

LATNQ

Function

Negative D-type latch with a single non-inverted output Q.



$$Q = \begin{cases} D & \text{if } CK = 0 \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
LATNQX1	3.36	5.46
LATNQX2	3.36	5.88
LATNQX3	3.36	5.88

Input pin capacitance

Cell	Capacitance [pF]	
	CK	D
LATNQX1	0.0016	0.0016
LATNQX2	0.0016	0.0016
LATNQX3	0.0016	0.0016

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	CK→Q	D→Q	CK→Q	D→Q
LATNQX1	0.5194	0.4476	0.4906	0.4170
LATNQX2	0.5353	0.4637	0.5046	0.4302
LATNQX3	0.5597	0.4881	0.5255	0.4511

Timing constraints

Cell	Setup time [ns]	Hold time [ns]	Min. pulse [ns]
	D↤CK	CK↦D	CK
LATNQX1	0.2452	0.0099	3.5156
LATNQX2	0.2535	0.0049	3.5156
LATNQX3	0.2541	0.0025	3.5156

Internal dynamic energy per switching event

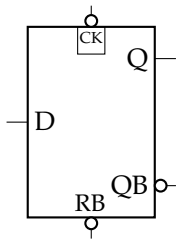
Cell	Energy [pJ]	
	CK→Q	D→Q
LATNQX1	0.01485	0.00990
LATNQX2	0.01720	0.01220
LATNQX3	0.01860	0.01365

Leakage power

Cell	Leakage [nW]
LATNQX1	0.1896
LATNQX2	0.2178
LATNQX3	0.2364

LATNR

Function



Negative D-type latch with asynchronous active-low reset RB and complementary outputs Q and QB.

$$Q = \begin{cases} 0 & \text{if RB} = 0 \\ D & \text{if RB} = 1 \text{ and CK} = 0 \\ Q & \text{otherwise} \end{cases}$$

Dimensions

Cell	Height [μm]	Width [μm]
LATNRX1	3.36	7.14
LATNRX1P5	3.36	7.14
LATNRX2	3.36	7.14
LATNRX3	3.36	7.14

Input pin capacitance

Cell	Capacitance [pF]		
	CK	D	RB
LATNRX1	0.0016	0.0016	0.0021
LATNRX1P5	0.0016	0.0016	0.0021
LATNRX2	0.0016	0.0016	0.0021
LATNRX3	0.0016	0.0016	0.0021

Delay

Cell	Rising delay [ns]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATNRX1	0.5419	0.4940	0.4816	0.5697	0.5098	0.4814
LATNRX1P5	0.5589	0.5160	0.5020	0.5921	0.5400	0.5042
LATNRX2	0.5727	0.5340	0.5186	0.6117	0.5668	0.5234
LATNRX3	0.6115	0.5775	0.5610	0.6558	0.6190	0.5661

Cell	Falling delay [ns]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATNRX1	0.5247	0.4731	0.4333	0.5589	0.5054	0.4933
LATNRX1P5	0.5357	0.4910	0.4437	0.5793	0.5319	0.5184
LATNRX2	0.5491	0.5103	0.4557	0.6025	0.5604	0.5455
LATNRX3	0.5801	0.5479	0.4832	0.6447	0.6080	0.5917

Timing constraints

Cell	Setup/rec. time [ns]		Hold/rem. time [ns]		Min. pulse [ns]	
	D⇤CK	RB⇤CK	CK↦D	CK↦RB	CK	RB
LATNRX1	0.3061	0.2962	0.0123	-0.2098	3.5156	3.5156
LATNRX1P5	0.3604	0.3431	-0.0197	-0.2394	3.5156	3.5156
LATNRX2	0.4073	0.3875	-0.0444	-0.2592	3.5156	3.5156
LATNRX3	0.4665	0.4394	-0.0815	-0.2913	3.5156	3.5156

Internal dynamic energy per switching event

Cell	Energy [pJ]					
	CK→Q	D→Q	RB→Q	CK→QB	D→QB	RB→QB
LATNRX1	0.00915	0.00485	0.00535	0.00910	0.00485	0.00535
LATNRX1P5	0.01020	0.00590	0.00640	0.01020	0.00590	0.00640
LATNRX2	0.01130	0.00695	0.00750	0.01130	0.00695	0.00750
LATNRX3	0.01270	0.00835	0.00890	0.01270	0.00840	0.00890

Leakage power

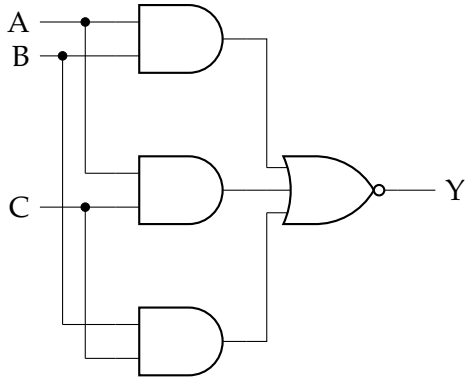
Cell	Leakage [nW]
LATNRX1	0.2054
LATNRX1P5	0.2339
LATNRX2	0.2618
LATNRX3	0.2989

MIN3

Function

3-input minority gate: output Y is high when fewer than two inputs are high.

$$Y = \overline{A \cdot B + A \cdot C + B \cdot C}$$



Dimensions

Cell	Height [µm]	Width [µm]
MIN3X1	3.36	2.94
MIN3X1P5	3.36	2.94
MIN3X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
MIN3X1	0.0030	0.0028	0.0016
MIN3X1P5	0.0046	0.0042	0.0025
MIN3X2	0.0062	0.0057	0.0032

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
MIN3X1	0.5599	0.5378	0.5493	0.5108	0.4956	0.4995
MIN3X1P5	0.5590	0.5335	0.5473	0.4973	0.4788	0.4867
MIN3X2	0.5512	0.5261	0.5402	0.4909	0.4746	0.4915

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
MIN3X1	0.00290	0.00280	0.00210
MIN3X1P5	0.00435	0.00410	0.00310
MIN3X2	0.00560	0.00535	0.00405

Leakage power

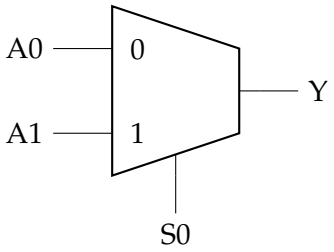
Cell	Leakage [nW]
MIN3X1	0.0595
MIN3X1P5	0.0999
MIN3X2	0.1386

MUX2

Function

2-to-1 multiplexer implemented using transmission gates.

$$Y = A0 \cdot \overline{S0} + A1 \cdot S0$$



Dimensions

Cell	Height [μm]	Width [μm]
MUX2X1	3.36	3.78
MUX2X2	3.36	3.78

Input pin capacitance

Cell	Capacitance [pF]		
	A0	A1	S0
MUX2X1	0.0016	0.0016	0.0031
MUX2X2	0.0016	0.0016	0.0031

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A0→Y	A1→Y	S0→Y	A0→Y	A1→Y	S0→Y
MUX2X1	0.3943	0.3884	0.3932	0.4004	0.3972	0.3912
MUX2X2	0.4366	0.4343	0.4307	0.4463	0.4436	0.4233

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A0→Y	A1→Y	S0→Y
MUX2X1	0.00430	0.00420	0.00695
MUX2X2	0.00620	0.00625	0.00905

Leakage power

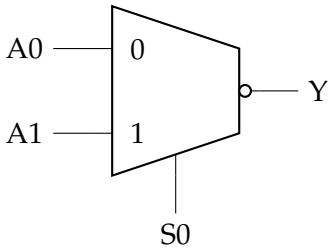
Cell	Leakage [nW]
MUX2X1	0.0991
MUX2X2	0.1273

MUXI2

Function

2-to-1 multiplexer with inverted output implemented using transmission gates.

$$Y = \overline{A0} \cdot \overline{S0} + \overline{A1} \cdot S0$$



Dimensions

Cell	Height [µm]	Width [µm]
MUXI2X1	3.36	3.36
MUXI2X2	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]		
	A0	A1	S0
MUXI2X1	0.0016	0.0016	0.0031
MUXI2X2	0.0033	0.0033	0.0047

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A0→Y	A1→Y	S0→Y	A0→Y	A1→Y	S0→Y
MUXI2X1	0.5467	0.5473	0.5252	0.4981	0.4972	0.4862
MUXI2X2	0.5505	0.5503	0.5289	0.4786	0.4787	0.4637

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A0→Y	A1→Y	S0→Y
MUXI2X1	0.00215	0.00210	0.00485
MUXI2X2	0.00440	0.00425	0.00825

Leakage power

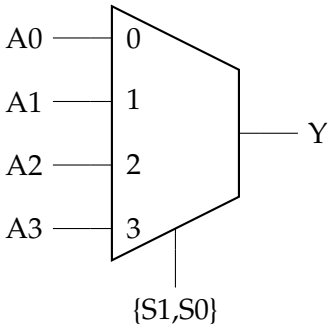
Cell	Leakage [nW]
MUXI2X1	0.0752
MUXI2X2	0.1404

MUX4

Function

4-to-1 multiplexer implemented using transmission gates.

$$Y = \begin{cases} A0 & \text{if } \{S1,S0\} = 00b \\ A1 & \text{if } \{S1,S0\} = 01b \\ A2 & \text{if } \{S1,S0\} = 10b \\ A3 & \text{if } \{S1,S0\} = 11b \end{cases}$$



Dimensions

Cell	Height [µm]	Width [µm]
MUX4X1	3.36	8.40
MUX4X1P5	3.36	8.40
MUX4X2	3.36	8.40
MUX4X3	3.36	8.40

Input pin capacitance

Cell	Capacitance [pF]					
	A0	A1	A2	A3	S0	S1
MUX4X1	0.0015	0.0015	0.0015	0.0015	0.0043	0.0033
MUX4X1P5	0.0015	0.0015	0.0015	0.0015	0.0043	0.0033
MUX4X2	0.0015	0.0015	0.0015	0.0015	0.0043	0.0033
MUX4X3	0.0015	0.0015	0.0015	0.0015	0.0043	0.0033

Delay

Cell	Rising delay [ns]					
	A0→Y	A1→Y	A2→Y	A3→Y	S0→Y	S1→Y
MUX4X1	0.4959	0.4937	0.4936	0.4909	0.4729	0.3666
MUX4X1P5	0.5165	0.5143	0.5134	0.5107	0.4962	0.3995
MUX4X2	0.5329	0.5307	0.5295	0.5268	0.5146	0.4250
MUX4X3	0.5723	0.5702	0.5690	0.5662	0.5563	0.4726

Cell	Falling delay [ns]					
	A0→Y	A1→Y	A2→Y	A3→Y	S0→Y	S1→Y
MUX4X1	0.5012	0.5002	0.5004	0.4987	0.4876	0.3729
MUX4X1P5	0.5217	0.5210	0.5201	0.5185	0.5112	0.4075
MUX4X2	0.5434	0.5425	0.5415	0.5400	0.5350	0.4387
MUX4X3	0.5837	0.5830	0.5821	0.5806	0.5783	0.4872

Internal dynamic energy per switching event

Cell	Energy [pJ]					
	A0→Y	A1→Y	A2→Y	A3→Y	S0→Y	S1→Y
MUX4X1	0.00650	0.00650	0.00645	0.00640	0.01205	0.00845
MUX4X1P5	0.00765	0.00760	0.00755	0.00750	0.01310	0.00950
MUX4X2	0.00870	0.00865	0.00860	0.00850	0.01415	0.01055
MUX4X3	0.01010	0.01005	0.00995	0.00985	0.01555	0.01195

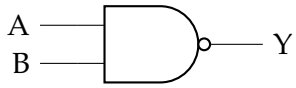
Leakage power

Cell	Leakage [nW]
MUX4X1	0.1891
MUX4X1P5	0.2034
MUX4X2	0.2173
MUX4X3	0.2359

NAND2

Function

2-input NAND.



$$Y = \overline{A \cdot B}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND2X1	3.36	1.68
NAND2X1P5	3.36	1.68
NAND2X2	3.36	1.68
NAND2X3	3.36	1.68
NAND2X4	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
NAND2X1	0.0016	0.0016
NAND2X1P5	0.0025	0.0024
NAND2X2	0.0033	0.0033
NAND2X3	0.0045	0.0044
NAND2X4	0.0066	0.0065

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
NAND2X1	0.4057	0.4130	0.4831	0.4749
NAND2X1P5	0.4013	0.4086	0.4709	0.4587
NAND2X2	0.3953	0.4024	0.4648	0.4506
NAND2X3	0.4151	0.4216	0.4817	0.4652
NAND2X4	0.3949	0.4027	0.4667	0.4534

Internal dynamic energy per switching event

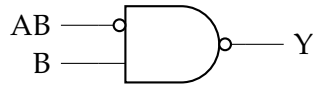
Cell	Energy [pJ]	
	A→Y	B→Y
NAND2X1	0.00135	0.00170
NAND2X1P5	0.00200	0.00250
NAND2X2	0.00270	0.00335
NAND2X3	0.00370	0.00450
NAND2X4	0.00525	0.00665

Leakage power

Cell	Leakage [nW]
NAND2X1	0.0297
NAND2X1P5	0.0484
NAND2X2	0.0668
NAND2X3	0.0914
NAND2X4	0.1337

NAND2B

Function



2-input NAND with one inverted input.

$$Y = \overline{\overline{AB} \cdot B}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND2BX1	3.36	2.10
NAND2BX1P5	3.36	2.10
NAND2BX2	3.36	2.10
NAND2BX3	3.36	2.10
NAND2BX4	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]	
	AB	B
NAND2BX1	0.0016	0.0016
NAND2BX1P5	0.0016	0.0024
NAND2BX2	0.0016	0.0033
NAND2BX3	0.0016	0.0045
NAND2BX4	0.0016	0.0066

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	AB→Y	B→Y	AB→Y	B→Y
NAND2BX1	0.3137	0.4130	0.3904	0.4759
NAND2BX1P5	0.3373	0.4087	0.4039	0.4605
NAND2BX2	0.3558	0.4029	0.4223	0.4533
NAND2BX3	0.3984	0.4152	0.4766	0.4819
NAND2BX4	0.4236	0.3961	0.5061	0.4654

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	AB→Y	B→Y
NAND2BX1	0.00375	0.00175
NAND2BX1P5	0.00495	0.00260
NAND2BX2	0.00615	0.00340
NAND2BX3	0.00855	0.00365
NAND2BX4	0.01195	0.00540

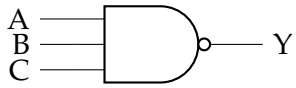
Leakage power

Cell	Leakage [nW]
NAND2BX1	0.0537
NAND2BX1P5	0.0723
NAND2BX2	0.0907
NAND2BX3	0.1153
NAND2BX4	0.1576

NAND3

Function

3-input NAND.



$$Y = \overline{A \cdot B \cdot C}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND3X1	3.36	2.10
NAND3X2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
NAND3X1	0.0016	0.0016	0.0016
NAND3X2	0.0033	0.0032	0.0033

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
NAND3X1	0.4172	0.4236	0.4104	0.5822	0.5651	0.5764
NAND3X2	0.4060	0.4123	0.4000	0.5497	0.5289	0.5503

Internal dynamic energy per switching event

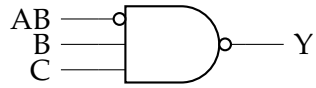
Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
NAND3X1	0.00210	0.00245	0.00175
NAND3X2	0.00405	0.00475	0.00340

Leakage power

Cell	Leakage [nW]
NAND3X1	0.0308
NAND3X2	0.0713

NAND3B

Function



3-input NAND with one inverted input.

$$Y = \overline{\overline{AB} \cdot B \cdot C}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND3BX1	3.36	2.94
NAND3BX2	3.36	2.94
NAND3BX3	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	AB	B	C
NAND3BX1	0.0016	0.0016	0.0016
NAND3BX2	0.0016	0.0033	0.0033
NAND3BX3	0.0016	0.0044	0.0044

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	AB→Y	B→Y	C→Y	AB→Y	B→Y	C→Y
NAND3BX1	0.3204	0.4184	0.4248	0.5038	0.5853	0.5689
NAND3BX2	0.3730	0.4061	0.4125	0.5349	0.5518	0.5309
NAND3BX3	0.4106	0.4247	0.4306	0.5799	0.5713	0.5500

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	AB→Y	B→Y	C→Y
NAND3BX1	0.00430	0.00220	0.00260
NAND3BX2	0.00715	0.00410	0.00480
NAND3BX3	0.00895	0.00550	0.00630

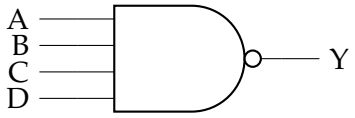
Leakage power

Cell	Leakage [nW]
NAND3BX1	0.0547
NAND3BX2	0.0952
NAND3BX3	0.1223

NAND4

Function

4-input NAND.



$$Y = \overline{A \cdot B \cdot C \cdot D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND4X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
NAND4X2	0.0033	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
NAND4X2	0.4208	0.4155	0.4098	0.4037	0.6255	0.6373	0.6489	0.6400

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
NAND4X2	0.00600	0.00535	0.00465	0.00390

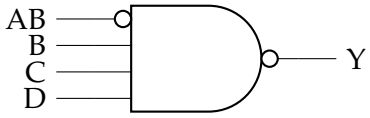
Leakage power

Cell	Leakage [nW]
NAND4X2	0.0745

NAND4B

Function

4-input NAND with one inverted input.



$$Y = \overline{\overline{AB} \cdot B \cdot C \cdot D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND4BX1	3.36	3.36
NAND4BX2	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]			
	AB	B	C	D
NAND4BX1	0.0016	0.0016	0.0016	0.0016
NAND4BX2	0.0016	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	B→Y	C→Y	D→Y	AB→Y	B→Y	C→Y	D→Y
NAND4BX1	0.3211	0.4204	0.4267	0.4320	0.6120	0.6919	0.6846	0.6764
NAND4BX2	0.3616	0.4099	0.4159	0.4220	0.6209	0.6523	0.6411	0.6300

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	B→Y	C→Y	D→Y
NAND4BX1	0.00445	0.00235	0.00280	0.00310
NAND4BX2	0.00745	0.00470	0.00545	0.00610

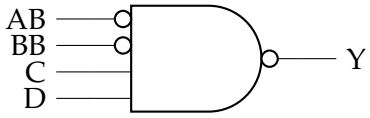
Leakage power

Cell	Leakage [nW]
NAND4BX1	0.0553
NAND4BX2	0.0984

NAND4BB

Function

A 4-input NAND with two negated inputs.



$$Y = \overline{\overline{AB} \cdot \overline{BB} \cdot C \cdot D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NAND4BBX1	3.36	4.20
NAND4BBX2	3.36	4.20

Input pin capacitance

Cell	Capacitance [pF]			
	AB	BB	C	D
NAND4BBX1	0.0016	0.0016	0.0016	0.0016
NAND4BBX2	0.0016	0.0016	0.0033	0.0032

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	BB→Y	C→Y	D→Y	AB→Y	BB→Y	C→Y	D→Y
NAND4BBX1	0.3255	0.3261	0.4276	0.4329	0.6198	0.6396	0.6873	0.6788
NAND4BBX2	0.3634	0.3657	0.4154	0.4215	0.6232	0.6458	0.6417	0.6294

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	BB→Y	C→Y	D→Y
NAND4BBX1	0.00470	0.00495	0.00290	0.00325
NAND4BBX2	0.00750	0.00815	0.00545	0.00615

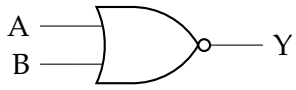
Leakage power

Cell	Leakage [nW]
NAND4BBX1	0.0792
NAND4BBX2	0.1223

NOR2

Function

2-input NOR.



$$Y = \overline{A + B}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR2X1	3.36	1.68
NOR2X2	3.36	1.68
NOR2X3	3.36	1.68
NOR2X4	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
NOR2X1	0.0016	0.0016
NOR2X2	0.0033	0.0033
NOR2X3	0.0045	0.0044
NOR2X4	0.0066	0.0065

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
NOR2X1	0.5331	0.5136	0.3899	0.3978
NOR2X2	0.5250	0.5022	0.3784	0.3855
NOR2X3	0.5549	0.5325	0.3908	0.3972
NOR2X4	0.5250	0.5051	0.3777	0.3860

Internal dynamic energy per switching event

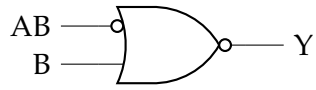
Cell	Energy [pJ]	
	A→Y	B→Y
NOR2X1	0.00130	0.00185
NOR2X2	0.00270	0.00360
NOR2X3	0.00350	0.00470
NOR2X4	0.00515	0.00735

Leakage power

Cell	Leakage [nW]
NOR2X1	0.0299
NOR2X2	0.0681
NOR2X3	0.0934
NOR2X4	0.1363

NOR2B

Function



2-input NOR with one inverted input.

$$Y = \overline{\overline{AB} + B}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR2BX1	3.36	2.10
NOR2BX2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]	
	AB	B
NOR2BX1	0.0016	0.0016
NOR2BX2	0.0016	0.0033

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	AB→Y	B→Y	AB→Y	B→Y
NOR2BX1	0.4523	0.5131	0.2939	0.3974
NOR2BX2	0.4910	0.5031	0.3380	0.3859

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	AB→Y	B→Y
NOR2BX1	0.00375	0.00180
NOR2BX2	0.00605	0.00355

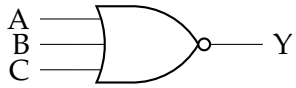
Leakage power

Cell	Leakage [nW]
NOR2BX1	0.0538
NOR2BX2	0.0920

NOR3

Function

3-input NOR.



$$Y = \overline{A + B + C}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR3X1	3.36	2.10
NOR3X2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
NOR3X1	0.0016	0.0016	0.0016
NOR3X2	0.0032	0.0033	0.0033

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
NOR3X1	0.6474	0.6669	0.6653	0.4073	0.4018	0.3932
NOR3X2	0.6331	0.6561	0.6577	0.3944	0.3892	0.3815

Internal dynamic energy per switching event

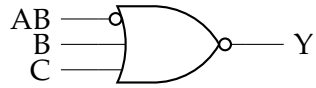
Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
NOR3X1	0.00265	0.00215	0.00160
NOR3X2	0.00515	0.00410	0.00320

Leakage power

Cell	Leakage [nW]
NOR3X1	0.0316
NOR3X2	0.0753

NOR3B

Function



3-input NOR with one inverted input.

$$Y = \overline{\overline{AB} + B + C}$$

Dimensions

Cell	Height [μm]	Width [μm]
NOR3BX1	3.36	2.52
NOR3BX1P5	3.36	2.52
NOR3BX2	3.36	2.94
NOR3BX3	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]		
	AB	B	C
NOR3BX1	0.0016	0.0017	0.0017
NOR3BX1P5	0.0025	0.0024	0.0024
NOR3BX2	0.0016	0.0033	0.0033
NOR3BX3	0.0016	0.0044	0.0044

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	AB→Y	B→Y	C→Y	AB→Y	B→Y	C→Y
NOR3BX1	0.6022	0.6625	0.6434	0.2996	0.4002	0.4054
NOR3BX1P5	0.6016	0.6641	0.6422	0.2879	0.3935	0.3984
NOR3BX2	0.6444	0.6583	0.6347	0.3459	0.3898	0.3944
NOR3BX3	0.7117	0.7046	0.6809	0.3813	0.4017	0.4055

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	AB→Y	B→Y	C→Y
NOR3BX1	0.00395	0.00195	0.00250
NOR3BX1P5	0.00605	0.00305	0.00380
NOR3BX2	0.00680	0.00415	0.00515
NOR3BX3	0.00850	0.00550	0.00680

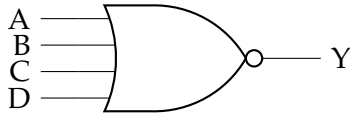
Leakage power

Cell	Leakage [nW]
NOR3BX1	0.0555
NOR3BX1P5	0.0920
NOR3BX2	0.0992
NOR3BX3	0.1281

NOR4

Function

4-input NOR.



$$Y = \overline{A + B + C + D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR4X1	3.36	2.94
NOR4X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
NOR4X1	0.0016	0.0016	0.0016	0.0017
NOR4X2	0.0032	0.0032	0.0032	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
NOR4X1	0.8153	0.8217	0.8254	0.8047	0.4155	0.4115	0.4052	0.3966
NOR4X2	0.7966	0.8065	0.8144	0.7982	0.4021	0.3980	0.3923	0.3846

Internal dynamic energy per switching event

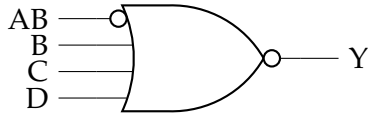
Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
NOR4X1	0.00355	0.00295	0.00240	0.00190
NOR4X2	0.00675	0.00570	0.00470	0.00370

Leakage power

Cell	Leakage [nW]
NOR4X1	0.0325
NOR4X2	0.0812

NOR4B

Function



4-input NOR with one negated input.

$$Y = \overline{\overline{AB} + B + C + D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR4BX1	3.36	3.36
NOR4BX2	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]			
	AB	B	C	D
NOR4BX1	0.0016	0.0016	0.0016	0.0016
NOR4BX2	0.0022	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	B→Y	C→Y	D→Y	AB→Y	B→Y	C→Y	D→Y
NOR4BX1	0.7577	0.8125	0.8187	0.8211	0.3023	0.4146	0.4105	0.4041
NOR4BX2	0.7709	0.7955	0.8055	0.8133	0.3160	0.4022	0.3981	0.3923

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	B→Y	C→Y	D→Y
NOR4BX1	0.00430	0.00345	0.00285	0.00230
NOR4BX2	0.00770	0.00660	0.00555	0.00455

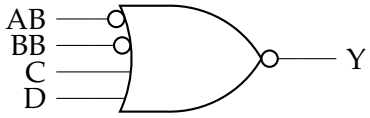
Leakage power

Cell	Leakage [nW]
NOR4BX1	0.0565
NOR4BX2	0.1146

NOR4BB

Function

A 4-input NOR with two negated inputs.



$$Y = \overline{\overline{AB} + \overline{BB} + C + D}$$

Dimensions

Cell	Height [µm]	Width [µm]
NOR4BBX1	3.36	3.78
NOR4BBX2	3.36	4.20

Input pin capacitance

Cell	Capacitance [pF]			
	AB	BB	C	D
NOR4BBX1	0.0016	0.0016	0.0016	0.0016
NOR4BBX2	0.0016	0.0016	0.0032	0.0032

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	BB→Y	C→Y	D→Y	AB→Y	BB→Y	C→Y	D→Y
NOR4BBX1	0.8233	0.7659	0.8203	0.8258	0.3089	0.3026	0.4114	0.4057
NOR4BBX2	0.8577	0.7945	0.8063	0.8141	0.3490	0.3439	0.3982	0.3924

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	BB→Y	C→Y	D→Y
NOR4BBX1	0.00595	0.00455	0.00300	0.00245
NOR4BBX2	0.01010	0.00725	0.00560	0.00460

Leakage power

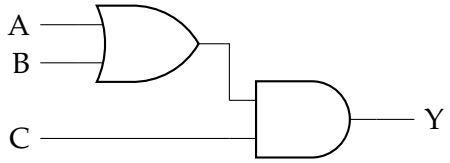
Cell	Leakage [nW]
NOR4BBX1	0.0804
NOR4BBX2	0.1290

OA21

Function

A 2-input OR followed by a 2-input AND.

$$Y = (A + B) \cdot C$$



Dimensions

Cell	Height [µm]	Width [µm]
OA21X2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
OA21X2	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
OA21X2	0.4379	0.4160	0.4468	0.4120	0.4208	0.3975

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
OA21X2	0.00640	0.00590	0.00690

Leakage power

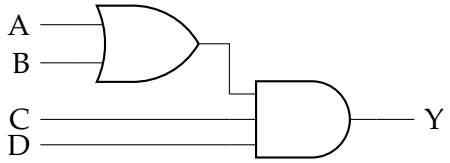
Cell	Leakage [nW]
OA21X2	0.0954

OA211

Function

A 2-input OR followed by a 3-input AND.

$$Y = (A + B) \cdot C \cdot D$$



Dimensions

Cell	Height [µm]	Width [µm]
OA211X1	3.36	2.94
OA211X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OA211X1	0.0016	0.0016	0.0016	0.0016
OA211X2	0.0016	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OA211X1	0.4541	0.4255	0.4737	0.4650	0.3872	0.3912	0.3803	0.3918
OA211X2	0.5023	0.4774	0.5175	0.5055	0.4293	0.4403	0.4080	0.4186

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OA211X1	0.00455	0.00410	0.00510	0.00540
OA211X2	0.00670	0.00620	0.00720	0.00760

Leakage power

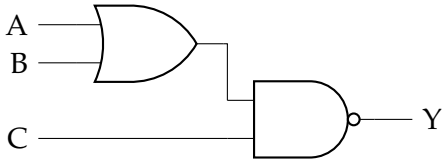
Cell	Leakage [nW]
OA211X1	0.0695
OA211X2	0.1026

OAI21

Function

A 2-input OR followed by a 2-input NAND.

$$Y = \overline{(A + B) \cdot C}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI21X1	3.36	2.10
OAI21X2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
OAI21X1	0.0015	0.0016	0.0016
OAI21X2	0.0031	0.0033	0.0033

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
OAI21X1	0.5304	0.5496	0.4074	0.4920	0.4807	0.4968
OAI21X2	0.5193	0.5410	0.3971	0.4670	0.4570	0.4776

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
OAI21X1	0.00245	0.00195	0.00195
OAI21X2	0.00470	0.00380	0.00370

Leakage power

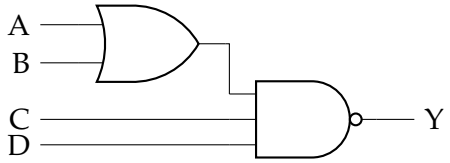
Cell	Leakage [nW]
OAI21X1	0.0412
OAI21X2	0.0949

OAI211

Function

A 2-input OR followed by a 3-input NAND.

$$Y = \overline{(A + B) \cdot C \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI211X1	3.36	2.52
OAI211X2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OAI211X1	0.0015	0.0016	0.0016	0.0016
OAI211X2	0.0031	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OAI211X1	0.5488	0.5669	0.4193	0.4128	0.5899	0.5740	0.6010	0.5943
OAI211X2	0.5346	0.5564	0.4080	0.4020	0.5520	0.5377	0.5665	0.5672

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OAI211X1	0.00330	0.00275	0.00270	0.00230
OAI211X2	0.00620	0.00530	0.00505	0.00440

Leakage power

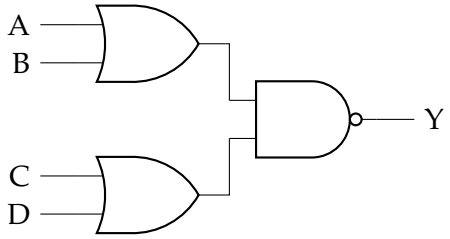
Cell	Leakage [nW]
OAI211X1	0.0429
OAI211X2	0.1014

OAI22

Function

Two 2-input ORs followed by a 2-input NAND.

$$Y = \overline{(A + B) \cdot (C + D)}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI22X1	3.36	2.94
OAI22X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OAI22X1	0.0015	0.0016	0.0016	0.0016
OAI22X2	0.0032	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OAI22X1	0.5412	0.5589	0.5405	0.5213	0.5111	0.5002	0.4994	0.5099
OAI22X2	0.5304	0.5515	0.5325	0.5096	0.4856	0.4762	0.4802	0.4896

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OAI22X1	0.00345	0.00275	0.00210	0.00260
OAI22X2	0.00650	0.00550	0.00410	0.00500

Leakage power

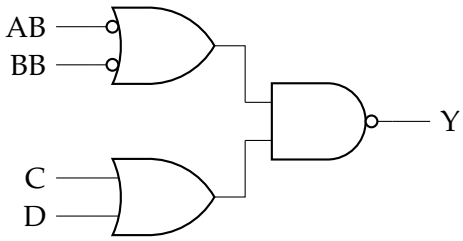
Cell	Leakage [nW]
OAI22X1	0.0505
OAI22X2	0.1163

OAI2BB2

Function

Two 2-input ORs, one of them with negated inputs, followed by a 2-input NAND.

$$Y = \overline{(\overline{AB} + \overline{BB}) \cdot (C + D)}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI2BB2X2	3.36	3.78

Input pin capacitance

Cell	Capacitance [pF]			
	AB	BB	C	D
OAI2BB2X2	0.0016	0.0016	0.0031	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	BB→Y	C→Y	D→Y	AB→Y	BB→Y	C→Y	D→Y
OAI2BB2X2	0.4132	0.4116	0.5193	0.5410	0.4625	0.4454	0.4694	0.4592

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	BB→Y	C→Y	D→Y
OAI2BB2X2	0.00785	0.00755	0.00475	0.00385

Leakage power

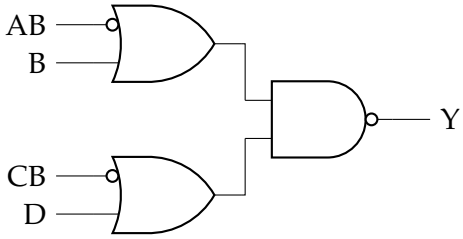
Cell	Leakage [nW]
OAI2BB2X2	0.1369

OAI2B2B

Function

Two 2-input ORs, each with an inverted input, followed by a 2-input NAND.

$$Y = \overline{(\overline{AB} + B) \cdot (\overline{CB} + D)}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI2B2BX1	3.36	4.20
OAI2B2BX1P5	3.36	4.20
OAI2B2BX2	3.36	4.20
OAI2B2BX3	3.36	4.62

Input pin capacitance

Cell	Capacitance [pF]			
	AB	B	CB	D
OAI2B2BX1	0.0016	0.0016	0.0016	0.0015
OAI2B2BX1P5	0.0016	0.0024	0.0016	0.0023
OAI2B2BX2	0.0016	0.0033	0.0016	0.0031
OAI2B2BX3	0.0016	0.0044	0.0016	0.0042

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	AB→Y	B→Y	CB→Y	D→Y	AB→Y	B→Y	CB→Y	D→Y
OAI2B2BX1	0.4606	0.5215	0.4800	0.5414	0.4074	0.5093	0.4276	0.5104
OAI2B2BX1P5	0.4854	0.5188	0.5052	0.5406	0.4199	0.4961	0.4411	0.4949
OAI2B2BX2	0.4990	0.5118	0.5190	0.5314	0.4366	0.4891	0.4569	0.4855
OAI2B2BX3	0.5514	0.5426	0.5703	0.5611	0.4792	0.5040	0.5006	0.4978

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	AB→Y	B→Y	CB→Y	D→Y
OAI2B2BX1	0.00470	0.00275	0.00580	0.00380
OAI2B2BX1P5	0.00635	0.00405	0.00810	0.00570
OAI2B2BX2	0.00780	0.00535	0.01010	0.00750
OAI2B2BX3	0.00990	0.00700	0.01295	0.00980

Leakage power

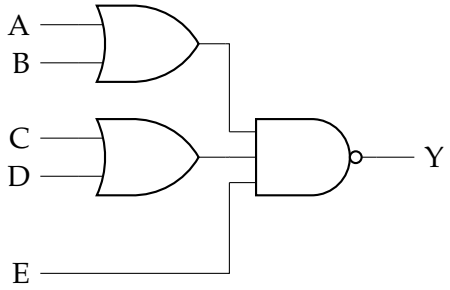
Cell	Leakage [nW]
OAI2B2BX1	0.0983
OAI2B2BX1P5	0.1316
OAI2B2BX2	0.1641
OAI2B2BX3	0.2077

OAI221

Function

Two 2-input ORs followed by a 3-input NAND.

$$Y = \overline{(A + B) \cdot (C + D) \cdot E}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI221X1	3.36	3.36
OAI221X2	3.36	3.36

Input pin capacitance

Cell	Capacitance [pF]				
	A	B	C	D	E
OAI221X1	0.0016	0.0015	0.0016	0.0016	0.0016
OAI221X2	0.0033	0.0031	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				
	A→Y	B→Y	C→Y	D→Y	E→Y
OAI221X1	0.5745	0.5567	0.5471	0.5283	0.4248
OAI221X2	0.5663	0.5458	0.5388	0.5167	0.4151

Cell	Falling delay [ns]				
	A→Y	B→Y	C→Y	D→Y	E→Y
OAI221X1	0.5998	0.6128	0.5953	0.6108	0.6263
OAI221X2	0.5625	0.5749	0.5681	0.5818	0.5897

Internal dynamic energy per switching event

Cell	Energy [pJ]				
	A→Y	B→Y	C→Y	D→Y	E→Y
OAI221X1	0.00365	0.00420	0.00250	0.00300	0.00350
OAI221X2	0.00715	0.00815	0.00480	0.00575	0.00675

Leakage power

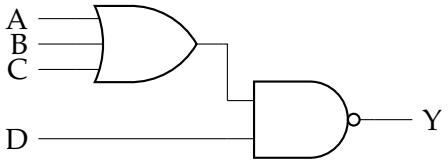
Cell	Leakage [nW]
OAI221X1	0.0526
OAI221X2	0.1227

OAI31

Function

A 3-input OR followed by a 2-input NAND.

$$Y = \overline{(A + B + C) \cdot D}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAI31X1	3.36	2.94
OAI31X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OAI31X1	0.0015	0.0015	0.0016	0.0016
OAI31X2	0.0031	0.0031	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OAI31X1	0.6689	0.6884	0.6861	0.4065	0.5039	0.4950	0.4837	0.5061
OAI31X2	0.6558	0.6790	0.6801	0.3970	0.4784	0.4713	0.4612	0.4862

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OAI31X1	0.00315	0.00255	0.00200	0.00235
OAI31X2	0.00615	0.00510	0.00410	0.00465

Leakage power

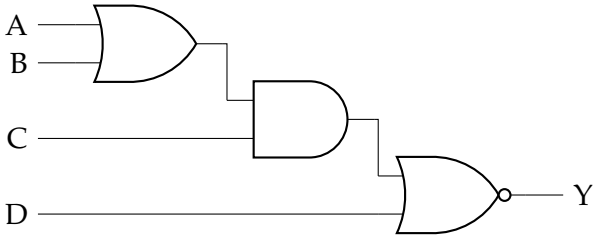
Cell	Leakage [nW]
OAI31X1	0.0479
OAI31X2	0.1132

OAOI211

Function

A 2-input OR followed by a 2-input AND followed by a 2-input NOR.

$$Y = \overline{(A + B) \cdot C + D}$$



Dimensions

Cell	Height [µm]	Width [µm]
OAOI211X1	3.36	2.94
OAOI211X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OAOI211X1	0.0016	0.0016	0.0016	0.0016
OAOI211X2	0.0033	0.0033	0.0033	0.0033

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OAOI211X1	0.6750	0.6723	0.5582	0.6635	0.5023	0.4910	0.4989	0.4169
OAOI211X2	0.6633	0.6643	0.5511	0.6499	0.4832	0.4722	0.4740	0.4040

Internal dynamic energy per switching event

Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OAOI211X1	0.00240	0.00180	0.00280	0.00380
OAOI211X2	0.00460	0.00365	0.00560	0.00760

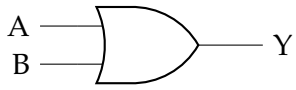
Leakage power

Cell	Leakage [nW]
OAOI211X1	0.0425
OAOI211X2	0.0997

OR2

Function

2-input OR.



$$Y = A + B$$

Dimensions

Cell	Height [µm]	Width [µm]
OR2X1	3.36	2.10
OR2X2	3.36	2.10
OR2X3	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
OR2X1	0.0016	0.0016
OR2X2	0.0016	0.0016
OR2X3	0.0016	0.0016

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
OR2X1	0.3208	0.3442	0.3507	0.3543
OR2X2	0.3616	0.3814	0.4076	0.4027
OR2X3	0.3979	0.4158	0.4498	0.4415

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	B→Y
OR2X1	0.00360	0.00400
OR2X2	0.00565	0.00615
OR2X3	0.00700	0.00750

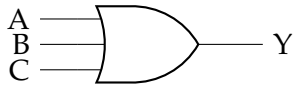
Leakage power

Cell	Leakage [nW]
OR2X1	0.0489
OR2X2	0.0732
OR2X3	0.0897

OR3

Function

3-input OR.



$$Y = A + B + C$$

Dimensions

Cell	Height [µm]	Width [µm]
OR3X1	3.36	2.52
OR3X2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
OR3X1	0.0016	0.0016	0.0016
OR3X2	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
OR3X1	0.3727	0.3578	0.3370	0.4272	0.4321	0.4210
OR3X2	0.4041	0.3905	0.3723	0.4773	0.4874	0.4823

Internal dynamic energy per switching event

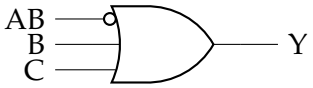
Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
OR3X1	0.00485	0.00430	0.00385
OR3X2	0.00695	0.00640	0.00595

Leakage power

Cell	Leakage [nW]
OR3X1	0.0481
OR3X2	0.0704

OR3B

Function



3-input OR with one inverted input.

$$Y = \overline{AB} + B + C$$

Dimensions

Cell	Height [µm]	Width [µm]
OR3BX1	3.36	2.52
OR3BX2	3.36	2.52

Input pin capacitance

Cell	Capacitance [pF]		
	AB	B	C
OR3BX1	0.0016	0.0016	0.0016
OR3BX2	0.0033	0.0016	0.0016

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	AB→Y	B→Y	C→Y	AB→Y	B→Y	C→Y
OR3BX1	0.4059	0.3289	0.3505	0.4836	0.4607	0.4599
OR3BX2	0.3952	0.3658	0.3848	0.4657	0.5015	0.4941

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	AB→Y	B→Y	C→Y
OR3BX1	0.00140	0.00430	0.00470
OR3BX2	0.00270	0.00690	0.00730

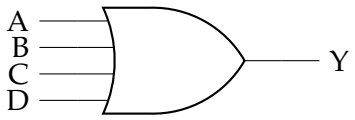
Leakage power

Cell	Leakage [nW]
OR3BX1	0.0526
OR3BX2	0.0817

OR4

Function

4-input OR.



$$Y = A + B + C + D$$

Dimensions

Cell	Height [µm]	Width [µm]
OR4X1	3.36	2.94
OR4X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]			
	A	B	C	D
OR4X1	0.0016	0.0016	0.0016	0.0016
OR4X2	0.0016	0.0016	0.0016	0.0016

Delay

Cell	Rising delay [ns]				Falling delay [ns]			
	A→Y	B→Y	C→Y	D→Y	A→Y	B→Y	C→Y	D→Y
OR4X1	0.3525	0.3722	0.3867	0.3974	0.4957	0.5145	0.5203	0.5164
OR4X2	0.3852	0.4019	0.4149	0.4245	0.5636	0.5782	0.5798	0.5745

Internal dynamic energy per switching event

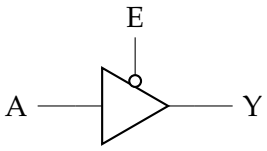
Cell	Energy [pJ]			
	A→Y	B→Y	C→Y	D→Y
OR4X1	0.00420	0.00465	0.00520	0.00575
OR4X2	0.00625	0.00675	0.00730	0.00780

Leakage power

Cell	Leakage [nW]
OR4X1	0.0479
OR4X2	0.0692

TBUF

Function



Tri-state buffer: The output Y is driven with the logical value of A when E is high, or is left in high impedance when E is low.

$$Y = \begin{cases} A & \text{if } E = 1 \\ Z & \text{if } E = 0 \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
TBUFX1	3.36	4.20
TBUFX2	3.36	4.20

Input pin capacitance

Cell	Capacitance [pF]	
	A	E
TBUFX1	0.0016	0.0032
TBUFX2	0.0016	0.0032

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	E→Y	A→Y	E→Y
TBUFX1	0.4144	0.3428	0.4046	0.3503
TBUFX2	0.4430	0.3659	0.4302	0.3565

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	E→Y
TBUFX1	0.00550	0.00400
TBUFX2	0.00740	0.00565

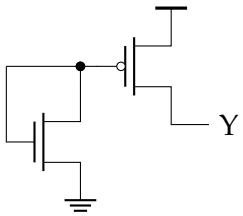
Leakage power

Cell	Leakage [nW]
TBUFX1	0.0852
TBUFX2	0.1019

TIEHI

Function

Provides a constant HI logic level without connecting directly to the VDD rail.



$$Y = 1$$

Dimensions

Cell	Height [µm]	Width [µm]
TIEHI	3.36	1.26

Leakage power

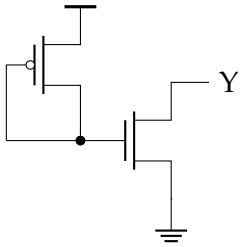
Cell	Leakage [nW]
TIEHI	0.0144

TIELO

Function

Provides a constant LO logic level without connecting directly to the VSS rail.

$$Y = 0$$



Dimensions

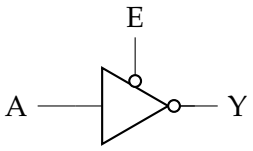
Cell	Height [µm]	Width [µm]
TIELO	3.36	1.26

Leakage power

Cell	Leakage [nW]
TIELO	0.0163

TINV

Function



Tri-state inverter: The output Y is driven with the logical complement of A when E is high, or is left in high impedance when E is low.

$$Y = \begin{cases} \overline{A} & \text{if } E = 1 \\ Z & \text{if } E = 0 \end{cases}$$

Dimensions

Cell	Height [µm]	Width [µm]
TINVX1	3.36	2.10
TINVX2	3.36	2.10

Input pin capacitance

Cell	Capacitance [pF]	
	A	E
TINVX1	0.0016	0.0023
TINVX2	0.0033	0.0030

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	E→Y	A→Y	E→Y
TINVX1	0.5296	0.4361	0.4798	0.4655
TINVX2	0.5226	0.4622	0.4616	0.4408

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	E→Y
TINVX1	0.00115	0.00235
TINVX2	0.00230	0.00430

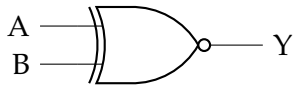
Leakage power

Cell	Leakage [nW]
TINVX1	0.0379
TINVX2	0.0599

XNOR2

Function

2-input XNOR.



$$Y = A \cdot B + \overline{A} \cdot \overline{B}$$

Dimensions

Cell	Height [μm]	Width [μm]
XNOR2X1	3.36	2.94
XNOR2X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
XNOR2X1	0.0028	0.0026
XNOR2X2	0.0045	0.0044

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
XNOR2X1	0.5162	0.5333	0.4983	0.4843
XNOR2X2	0.5194	0.5428	0.4632	0.4515

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	B→Y
XNOR2X1	0.00485	0.00470
XNOR2X2	0.00820	0.00815

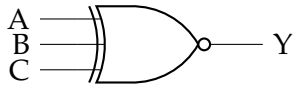
Leakage power

Cell	Leakage [nW]
XNOR2X1	0.0764
XNOR2X2	0.1332

XNOR3

Function

3-input XNOR.



$$Y = \overline{A \oplus B \oplus C}$$

Dimensions

Cell	Height [μm]	Width [μm]
XNOR3X1	3.36	6.30
XNOR3X2	3.36	6.30

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
XNOR3X1	0.0016	0.0032	0.0032
XNOR3X2	0.0016	0.0032	0.0032

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
XNOR3X1	0.5478	0.4918	0.3606	0.5625	0.5309	0.3804
XNOR3X2	0.5792	0.5270	0.4097	0.5978	0.5742	0.4428

Internal dynamic energy per switching event

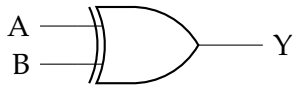
Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
XNOR3X1	0.01260	0.01145	0.00815
XNOR3X2	0.01480	0.01360	0.01020

Leakage power

Cell	Leakage [nW]
XNOR3X1	0.1888
XNOR3X2	0.2170

XOR2

Function



2-input XOR.

$$Y = A \cdot \overline{B} + \overline{A} \cdot B$$

Dimensions

Cell	Height [μm]	Width [μm]
XOR2X1	3.36	2.94
XOR2X2	3.36	2.94

Input pin capacitance

Cell	Capacitance [pF]	
	A	B
XOR2X1	0.0028	0.0026
XOR2X2	0.0045	0.0043

Delay

Cell	Rising delay [ns]		Falling delay [ns]	
	A→Y	B→Y	A→Y	B→Y
XOR2X1	0.5517	0.5365	0.4796	0.4874
XOR2X2	0.5412	0.5280	0.4536	0.4679

Internal dynamic energy per switching event

Cell	Energy [pJ]	
	A→Y	B→Y
XOR2X1	0.00535	0.00525
XOR2X2	0.00865	0.00875

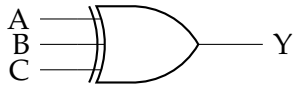
Leakage power

Cell	Leakage [nW]
XOR2X1	0.0718
XOR2X2	0.1240

XOR3

Function

3-input XOR.



$$Y = A \oplus B \oplus C$$

Dimensions

Cell	Height [µm]	Width [µm]
XOR3X1	3.36	6.30
XOR3X2	3.36	6.30

Input pin capacitance

Cell	Capacitance [pF]		
	A	B	C
XOR3X1	0.0016	0.0032	0.0032
XOR3X2	0.0016	0.0032	0.0032

Delay

Cell	Rising delay [ns]			Falling delay [ns]		
	A→Y	B→Y	C→Y	A→Y	B→Y	C→Y
XOR3X1	0.5442	0.4875	0.3723	0.5597	0.5269	0.3967
XOR3X2	0.5750	0.5225	0.4285	0.5956	0.5707	0.4081

Internal dynamic energy per switching event

Cell	Energy [pJ]		
	A→Y	B→Y	C→Y
XOR3X1	0.01245	0.01220	0.00830
XOR3X2	0.01455	0.01430	0.01035

Leakage power

Cell	Leakage [nW]
XOR3X1	0.1888
XOR3X2	0.2170